

Autonomic Nervous System

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Do.
Make.
Heal.
Innovate.

Reinvent the Future.

Session Objectives

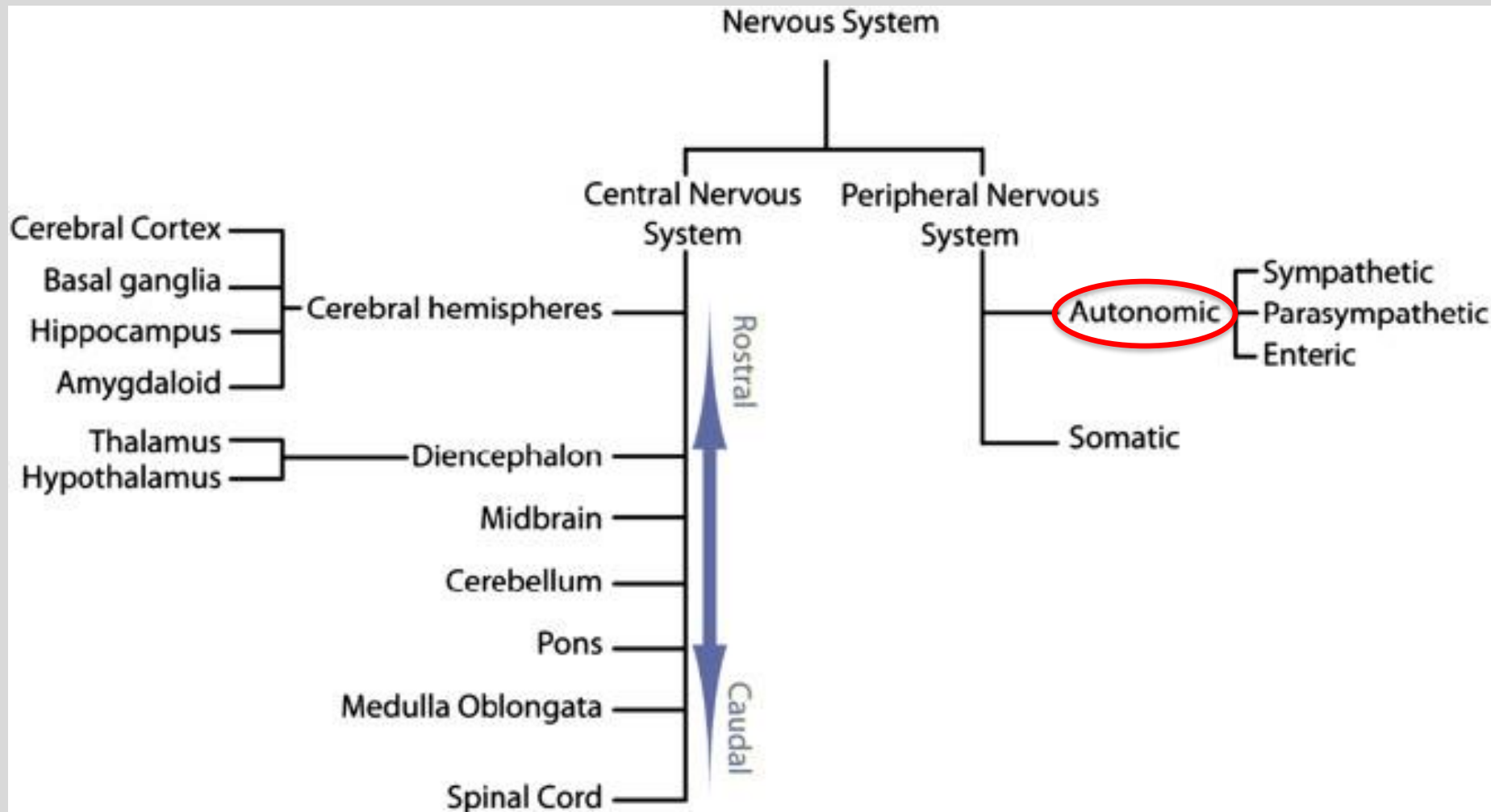
At the end of this lecture, students will be able to:

1. Describe the anatomy and neurotransmitters of the autonomic nervous system (ANS)
2. Depict the actions of the sympathetic and parasympathetic nervous system
3. Depict anatomy and physiology of the enteric nervous system and its neuromodulation
4. Comprehend the physiological basis of autonomic reflexes and several autonomic disorders

Organization of the nervous system

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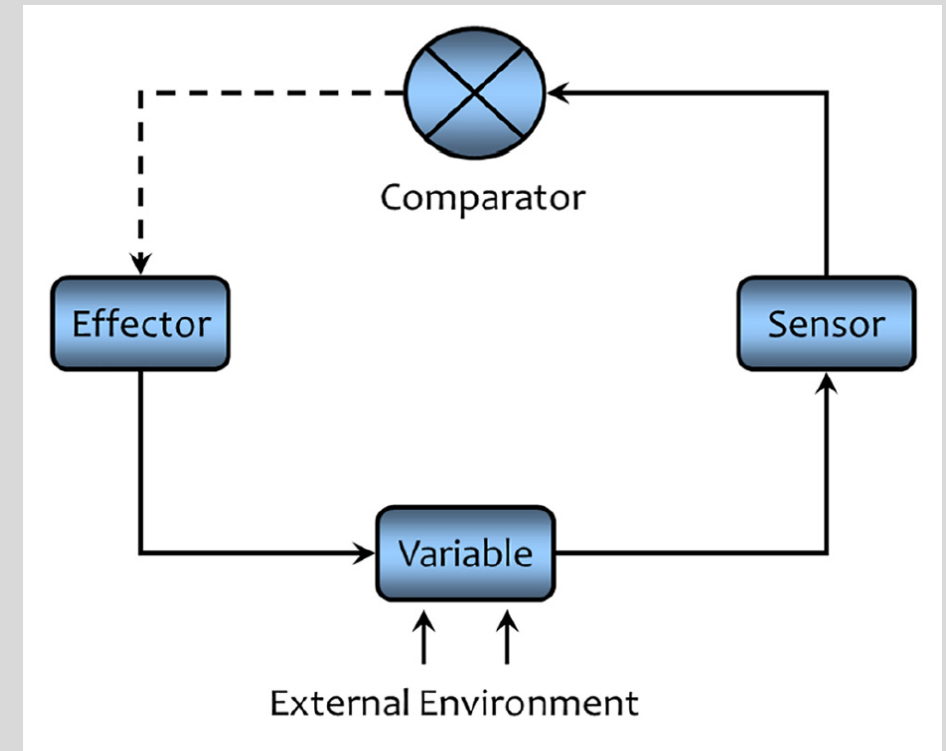
ANS and Homeostasis

“Autonomic” = self-rule/self-govern

Greek: auto = self; nomos = rule, law

“Homeostasis” = refers to any process that living things use to actively maintain fairly stable conditions necessary for survival.

*Greek: homeo = similar to;
stasis = standing still.*



—————▶ **Direct relationship**

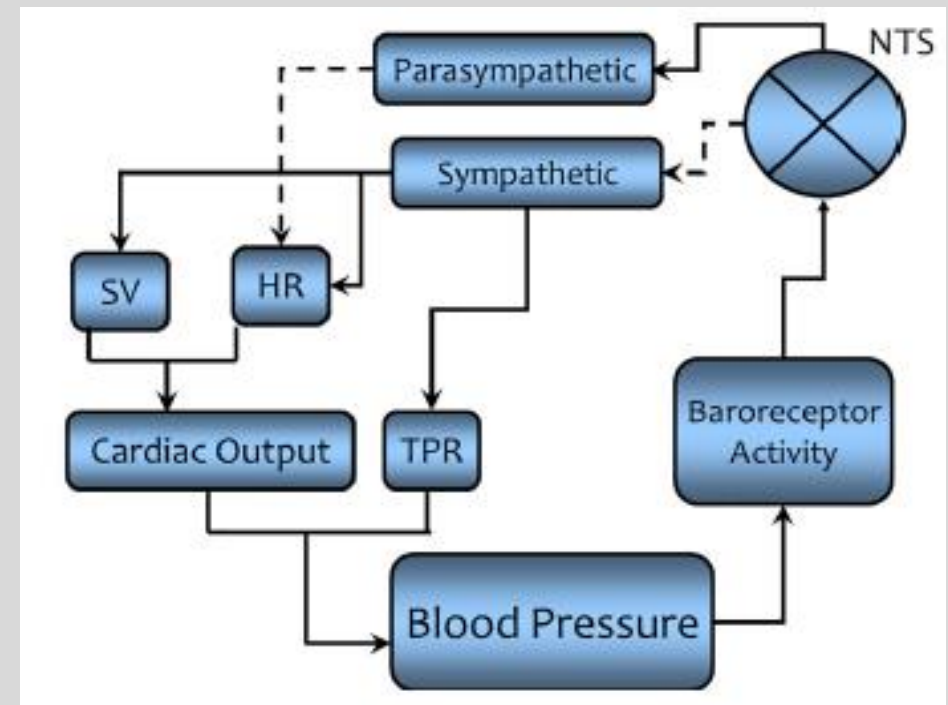
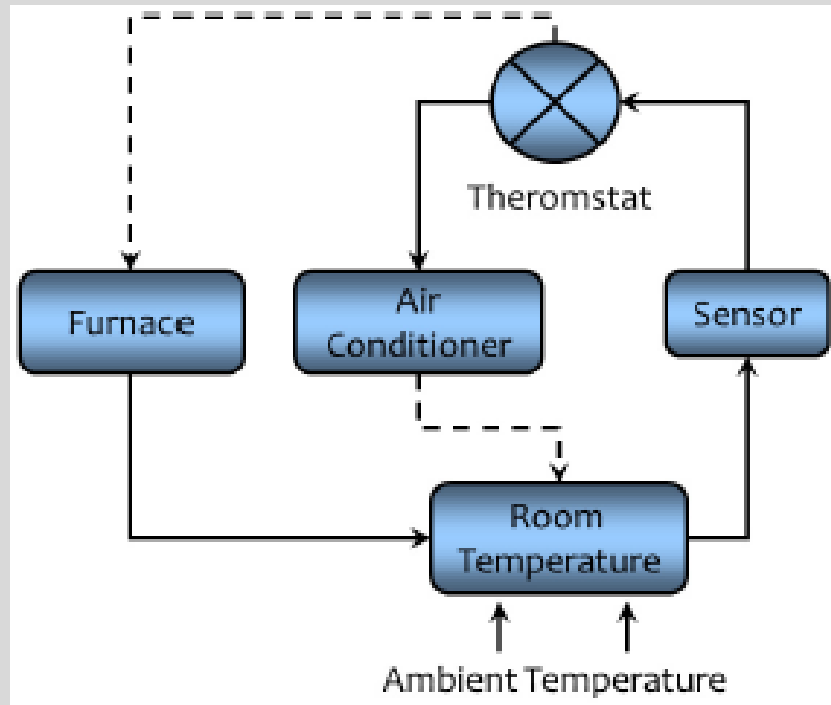
- - - - -▶ **Inverse relationship**

ANS and Homeostasis

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—————➔ **Direct relationship**
-----➔ **Inverse relationship**

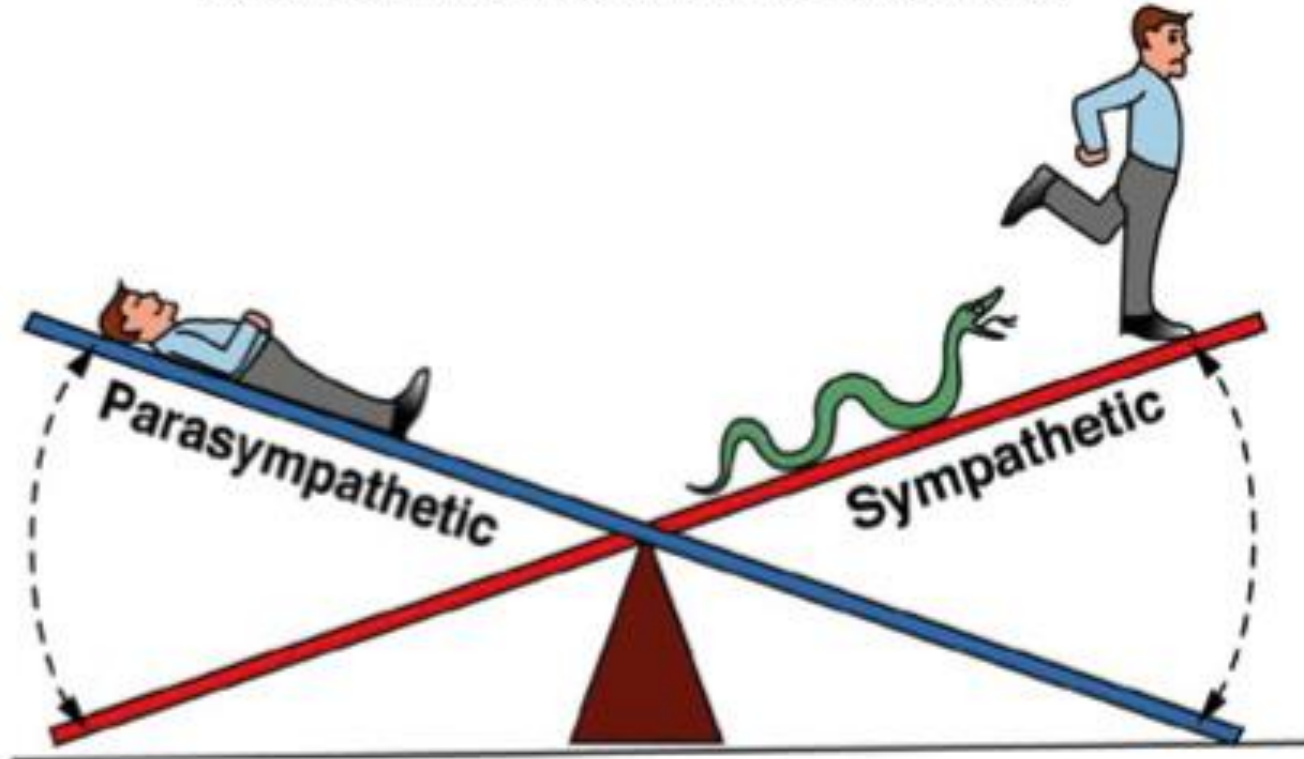


ANS and Homeostasis

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**Homeostasis is a Synergistic Balance
between the Autonomic Branches**



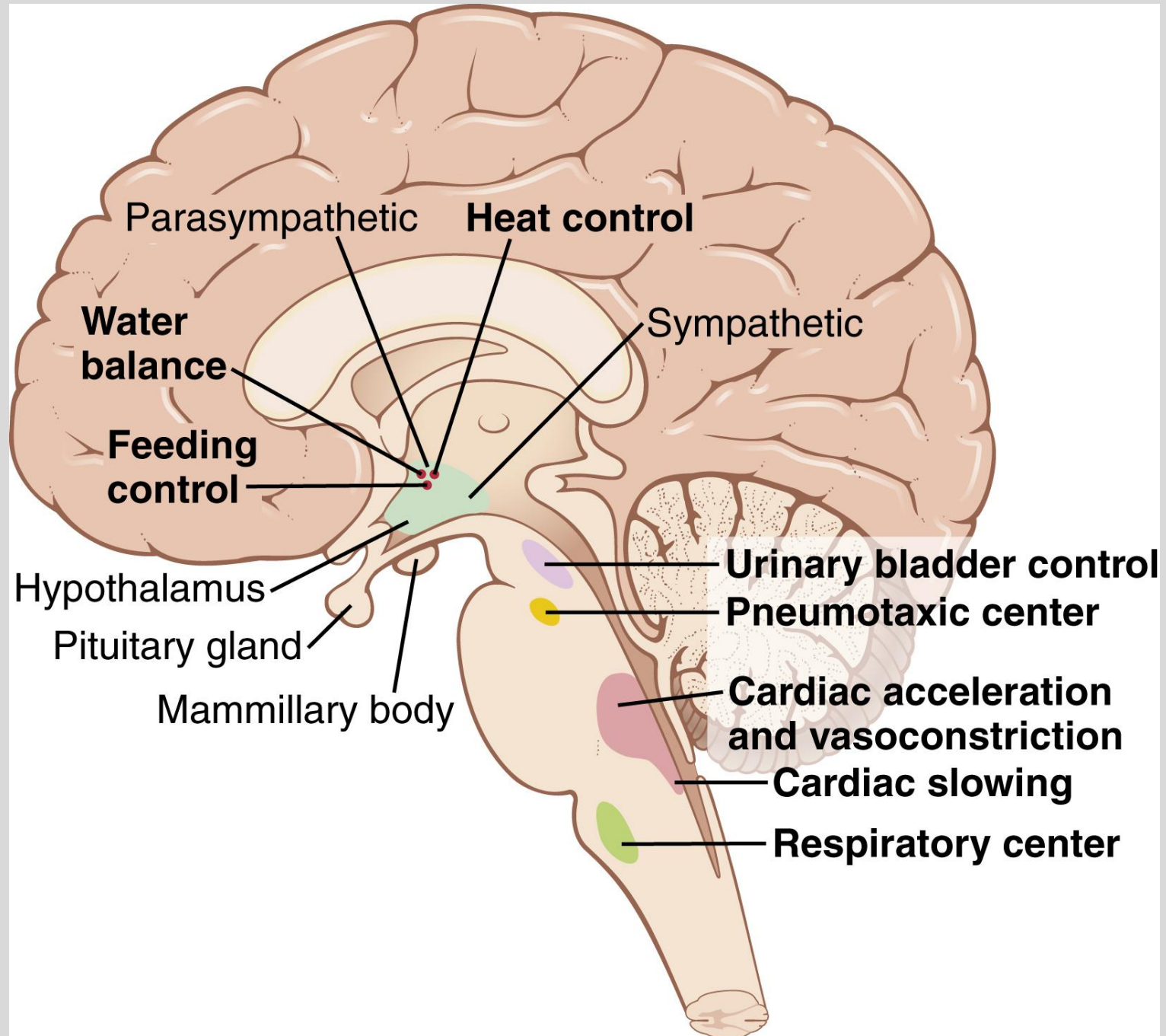
'Rest, Heal & Digest':
Parasympathetic
activity dominates.

'Fight or Flight':
Sympathetic
activity dominates.

<https://www.replenishyourlife.co.uk/post/the-role-of-the-autonomic-nervous-system-in-cfsme>

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Guyton and Hall Textbook of Medical
Physiology, Chapter 61, 763-775

Autonomic Innervation of Selected Body Structures

Structure	Sympathetic	Parasympathetic
eye - pupil (iris):	dilation	constriction
heart:	increased rate & force	decreased rate
lung bronchioles:	dilation	constriction
gut wall:	decreased motility	increased motility
gut sphincter:	constrict	relax
bladder detrusor:	relax	contraction
bladder-urethra smooth sphincter:	constrict	relax
penis etc.:	ejaculation	erection
gallbladder bile & duct:	relax	contract
salivary glands:	concentrated viscous saliva	abundant watery saliva
nasal & lacrimal glands:	vasoconstriction	abundant secretion
liver:	glucose production	---
kidney:	renin secretion	---
vessels within muscle:	dilation	---
cutaneous & visceral vessels:	constriction	---
piloerector muscles:	contraction	---
sweat glands (cholinergic):	secretion	---
adipose cells:	lipolysis	---
adrenal medulla:	epinephrine secretion	---

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“**P**oint and **S**hoot”

Erection: **P**arasympathetic

Ejaculation: **S**ympathetic

Dual innervation of many organs
— having a brake and an
accelerator provides more control

Overview of the ANS

Similarities and differences between Sympathetic and Parasympathetic

Target Tissues

Sympathetic

- Organs of head, neck, trunk, & external genitalia
- Adrenal medulla
- Sweat glands in skin
- Arrector muscles of hair
- *ALL* vascular smooth muscle

Parasympathetic

- Organs of head, neck, trunk, & external genitalia

- » Sympathetic system is distributed to essentially all tissues (because of vascular smooth muscle)
- » Parasympathetic system never reaches limbs or body wall (except for external genitalia)

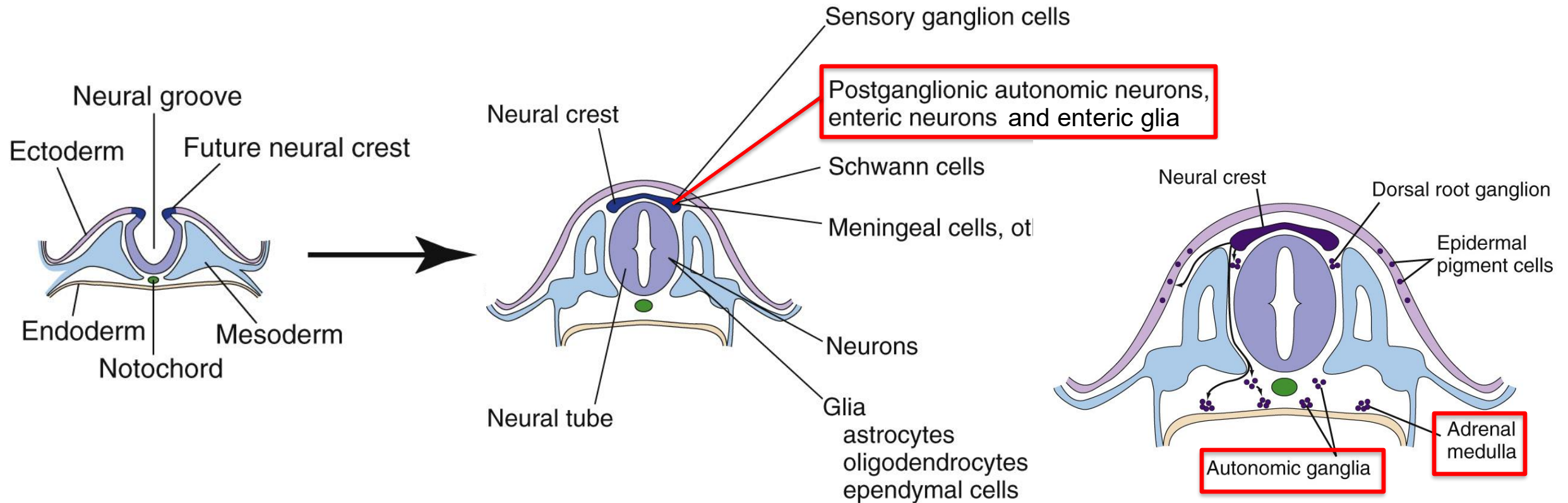
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ANS development

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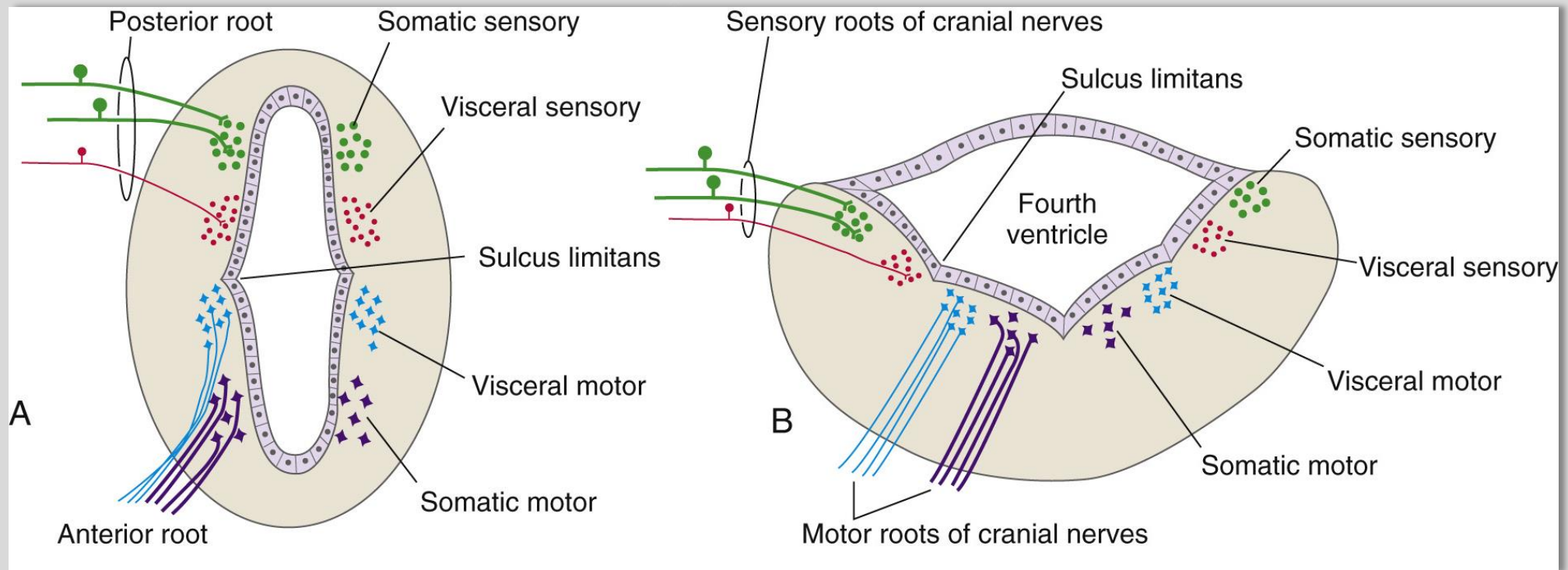
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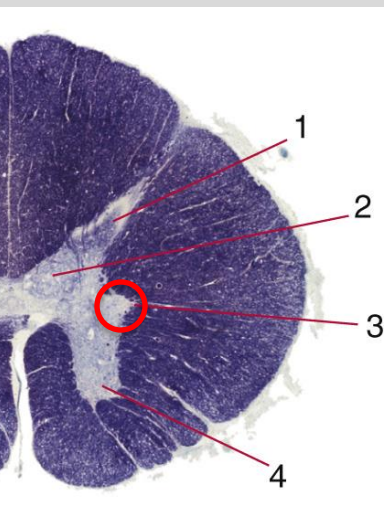


ANS development

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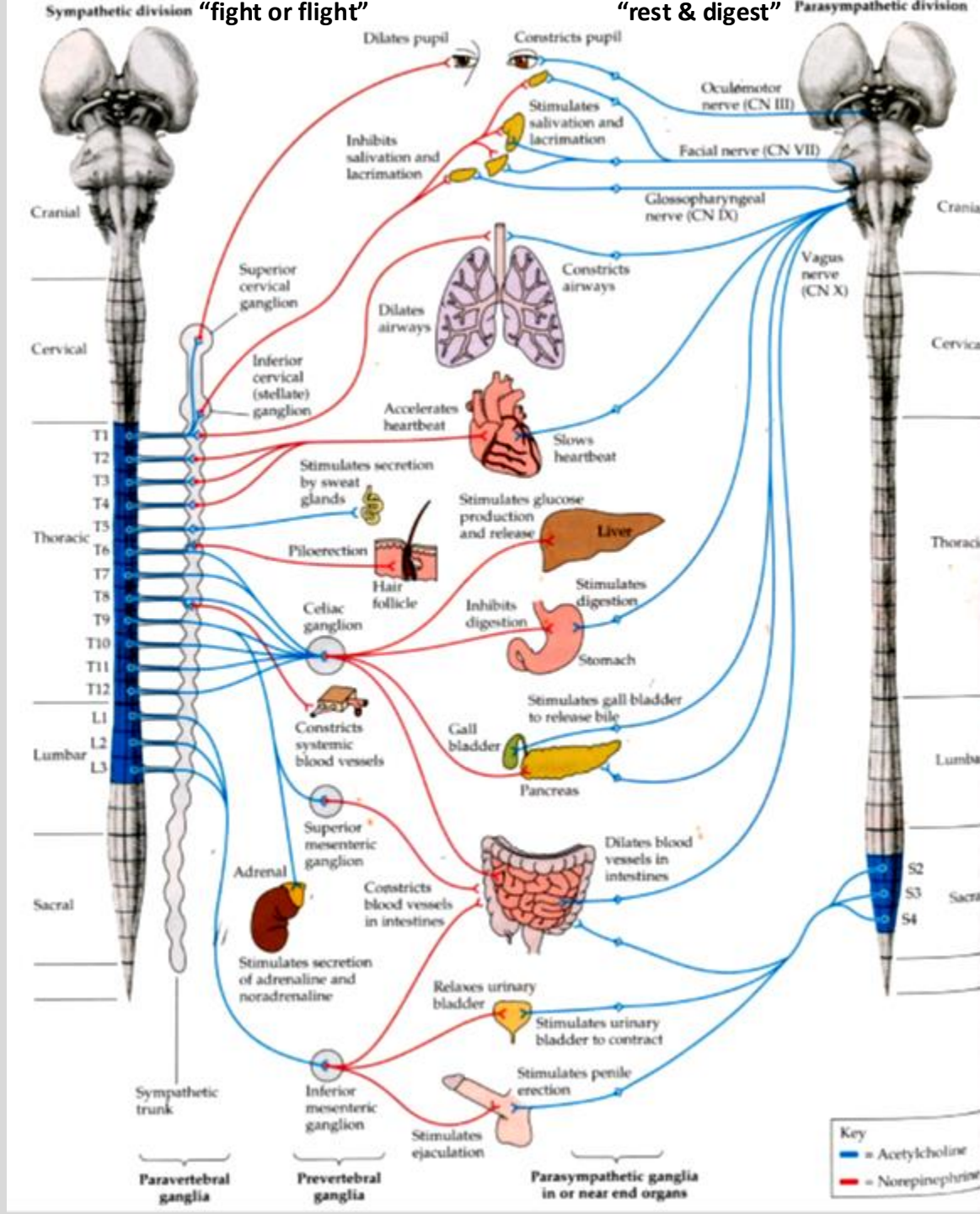
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Intermediolateral cell column

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Dorsal motor nucleus of Vagus n.

Intermediolateral cell column

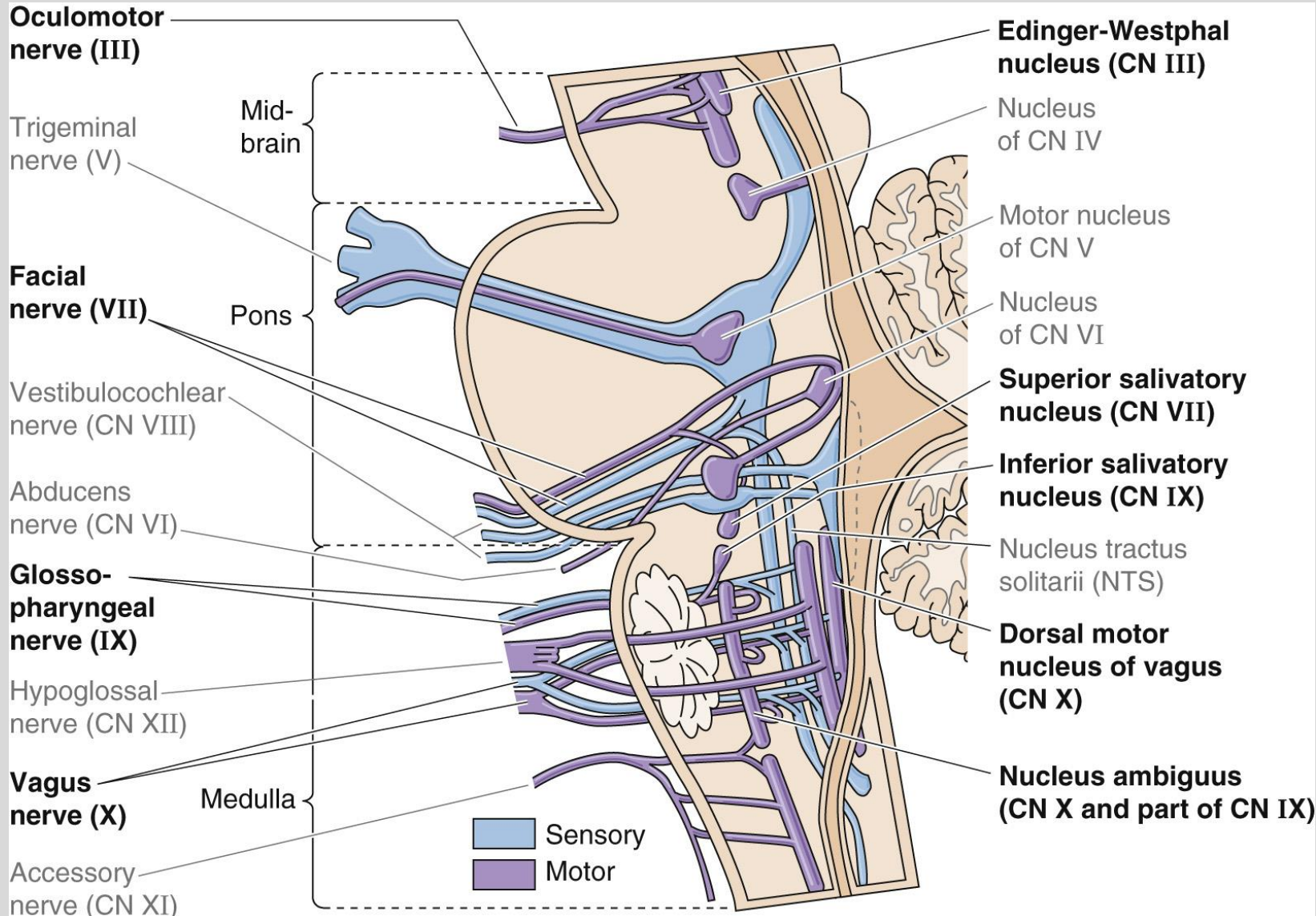
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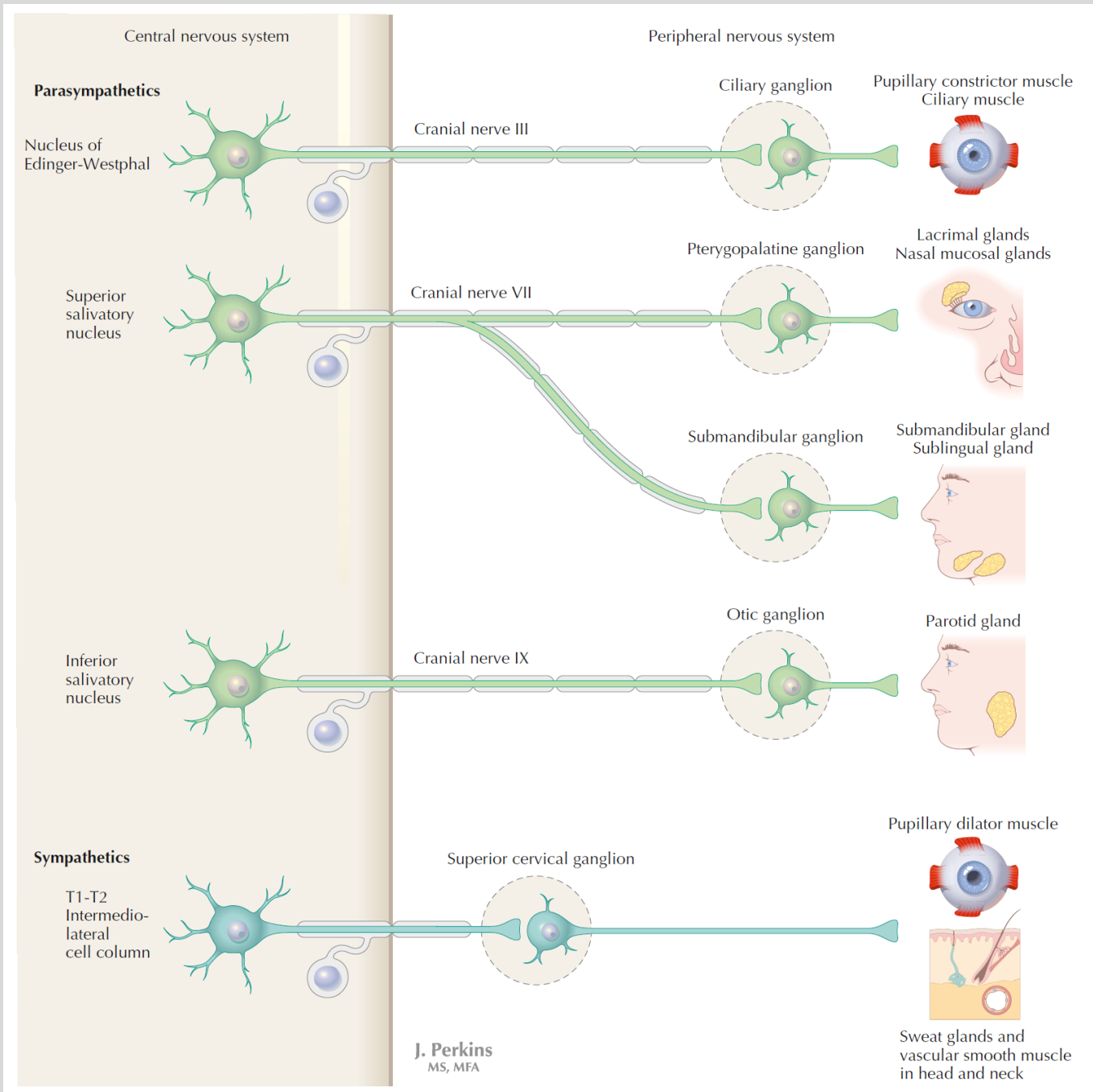


Supraspinal nuclei containing neurons that are part of the ANS

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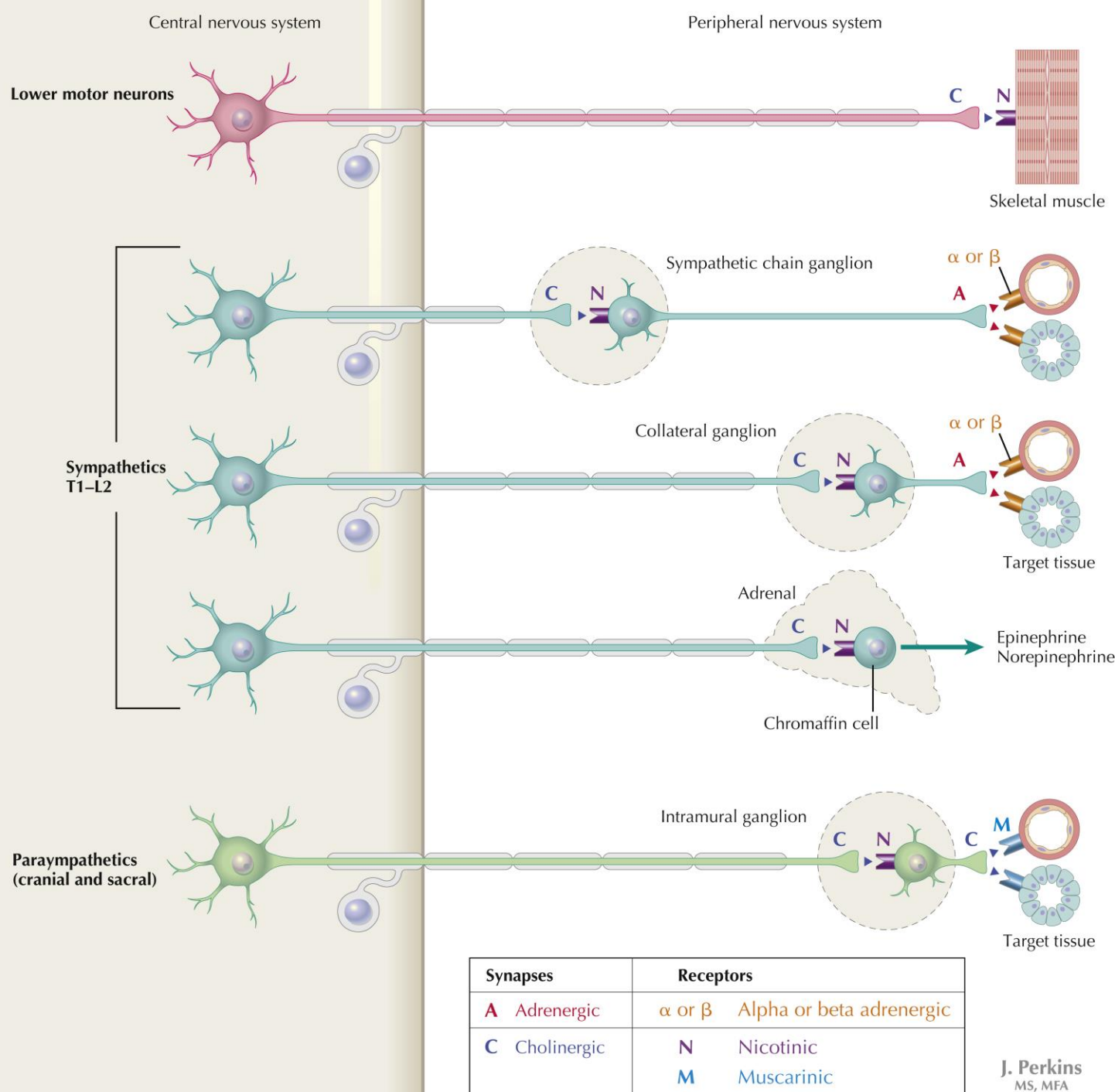
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Horner syndrome happens after sympathetic injury and it is the clinical triad of ipsilateral:

1. Ptosis (droopy eyelid),
2. Miosis (constricted pupil), and
3. facial anhidrosis (sometimes).

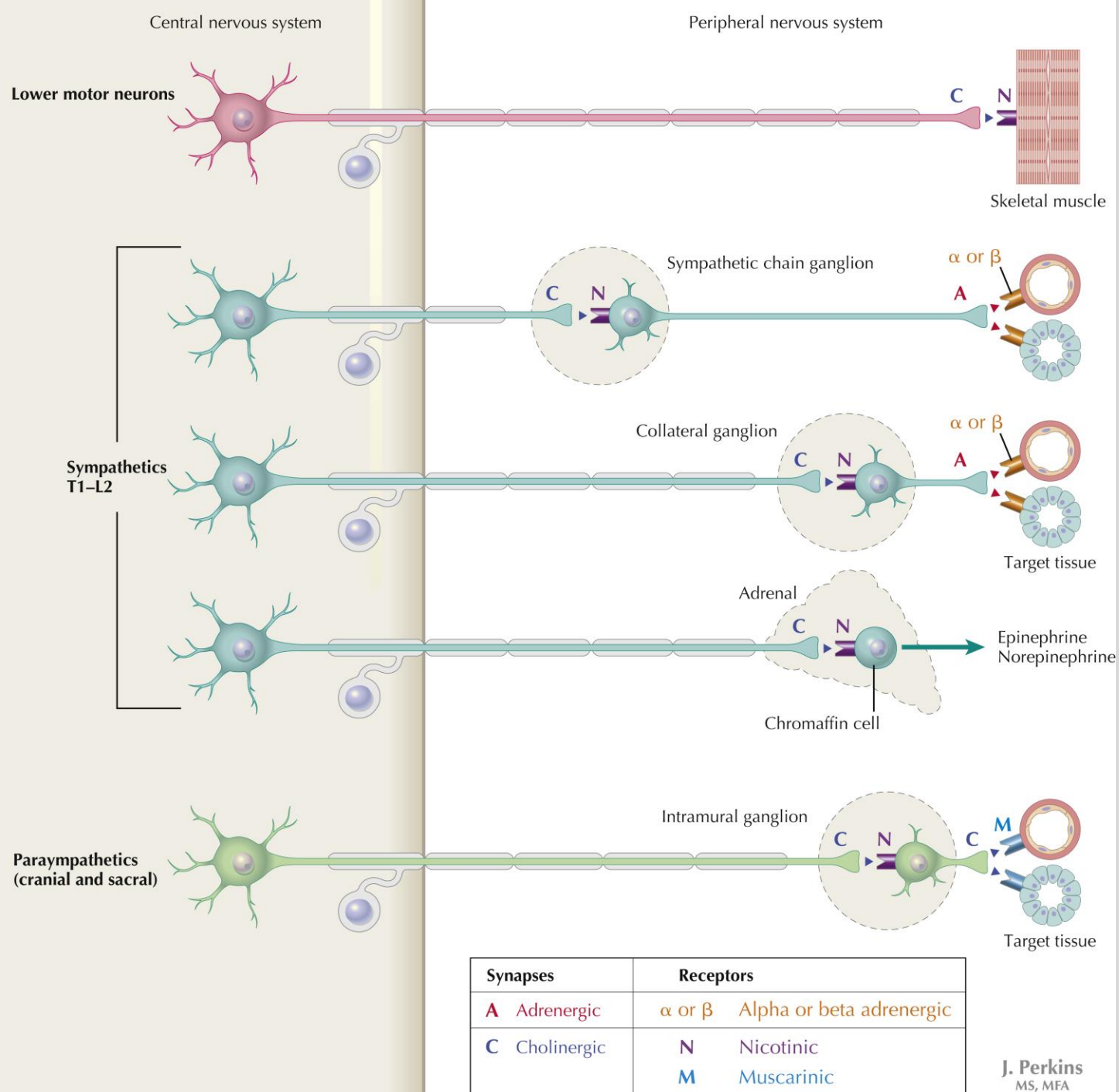


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Similarities between Sympathetic & Parasympathetic

- Both are visceromotor and autonomic
- Both involve 2 neurons that synapse in a peripheral ganglion
- Both have myelinated preganglionic and unmyelinated postganglionic neuron
- Both have cholinergic (release ACh) preganglionic neuron
- Both have nicotinic receptors on the postganglionic neuron
- Both have excitatory preganglionic neuron



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Differences between Sympathetic & Parasympathetic

- Preganglionic neuron body location in the CNS
 - Sympathetic is Thoraco-Lumbar
 - Parasympathetic is Cranio-Sacral
- Length of the pre- and post-ganglionic neurons:
 - Symp. has short pre- and long post-ganglionic
 - Parasymp. has long pre- and short post-ganglionic
- Postganglionic neurons:
 - Parasympathetic are cholinergic (release ACh)
 - Sympathetic are adrenergic (release NE) with **2 exceptions: sweat gland innervation** (release Ach) **and adrenal medulla** (release NE and Epi)
- The effect depends on the receptors expressed in the target tissue, i.e. could be excitatory or inhibitory

The effect depends on the receptors

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Autonomic Metabotropic Receptors

Muscarinic receptors:

Parasympathetic

Receptor	G-protein	Messenger	Effect	Example Functions
M ₁	G _q	Phospholipase C & IP ₃ increased	elevated Ca ⁺⁺	salivary gland & stomach secretion
M ₂	G _i	cAMP decreased	decreased Ca ⁺⁺ influx & increased K ⁺ efflux	decreased heart rate and force
M ₃	G _q	Phospholipase C & IP ₃ increased	elevated Ca ⁺⁺	constriction of vessels & bronchioles detrusor contraction (micturition)
M ₄	G _i	cAMP decreased	decreased Ca ⁺⁺ influx & increased K ⁺ efflux	inhibitory effects
M ₅	G _q	Phospholipase C & IP ₃ increased	elevated Ca ⁺⁺	present in CNS

Sweat gland secretion (Ach released by sympathetic postganglionic neurons)

Adrenergic receptors:

Sympathetic

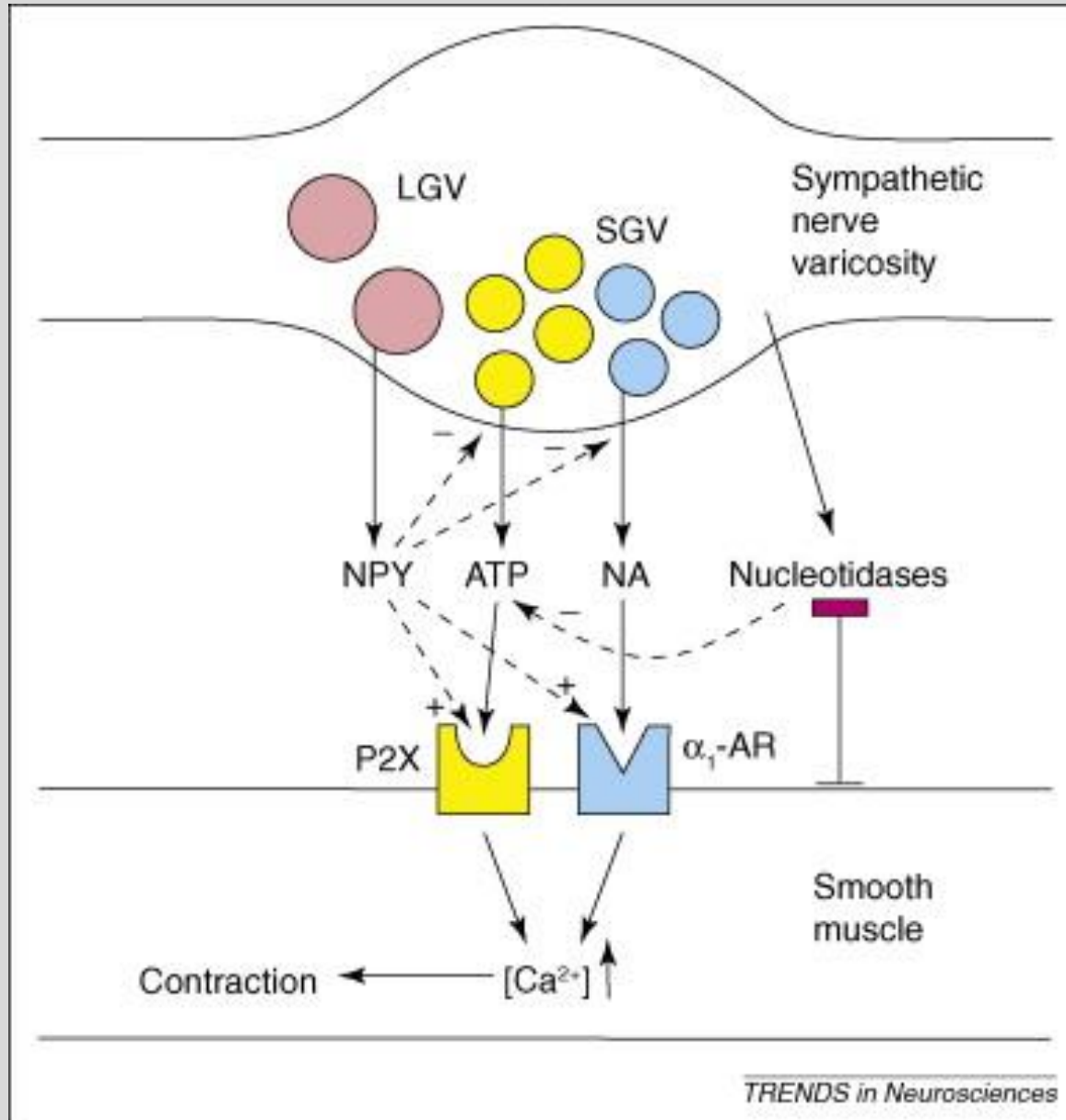
Receptor	G-protein	Messenger	Effect	Example Functions
α ₁	G _q	Phospholipase C & IP ₃ increased	elevated Ca ⁺⁺	cutaneous & GI vasoconstriction; urethral sphincter contraction
α ₂	G _i	cAMP decreased	decreased Ca ⁺⁺ influx	inhibition of neurotransmitter release
β ₁	G _s	cAMP increased	elevated Ca ⁺⁺	increased cardiac output (rate & force)
β ₂	G _s G _i	cAMP increased cAMP decreased	elevated Ca ⁺⁺ decreased Ca ⁺⁺ influx & increased K ⁺ efflux	constriction of gut sphincters muscle vessel dilation; detrusor relaxation
β ₃	G _s G _i	cAMP increased cAMP decreased	elevated Ca ⁺⁺ decreased Ca ⁺⁺ influx & increased K ⁺ efflux	adipose tissue lipolysis detrusor relaxation

<http://vanat.cvm.umn.edu/ans/pages/Physiol.html>

Co-transmission in the ANS

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LGV, large granular vesicles
SGV, small granular vesicles

NPY, Neuropeptide Y
ATP, Adenosine triphosphate
NA, noradrenaline

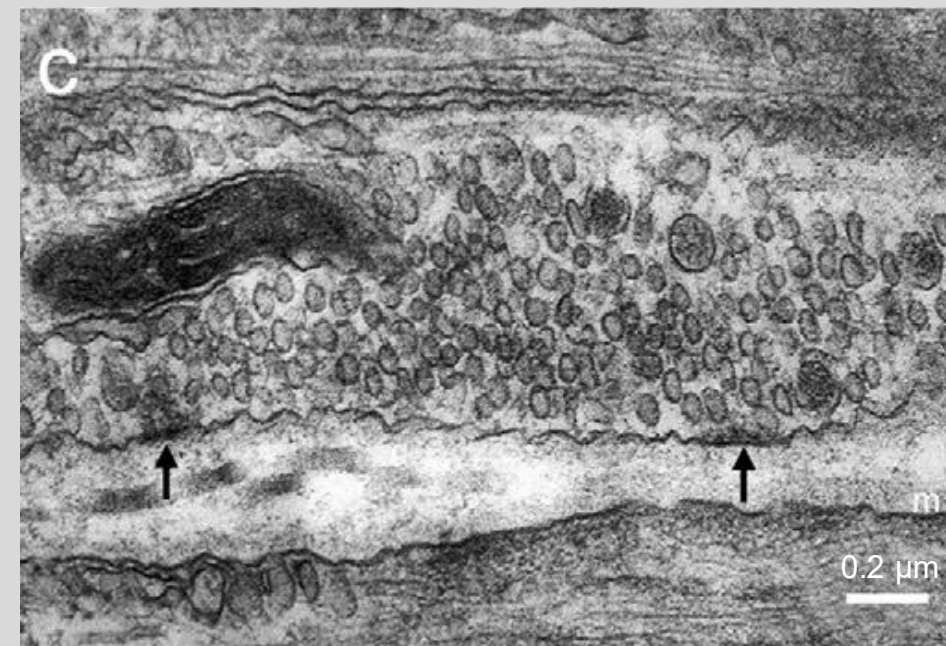
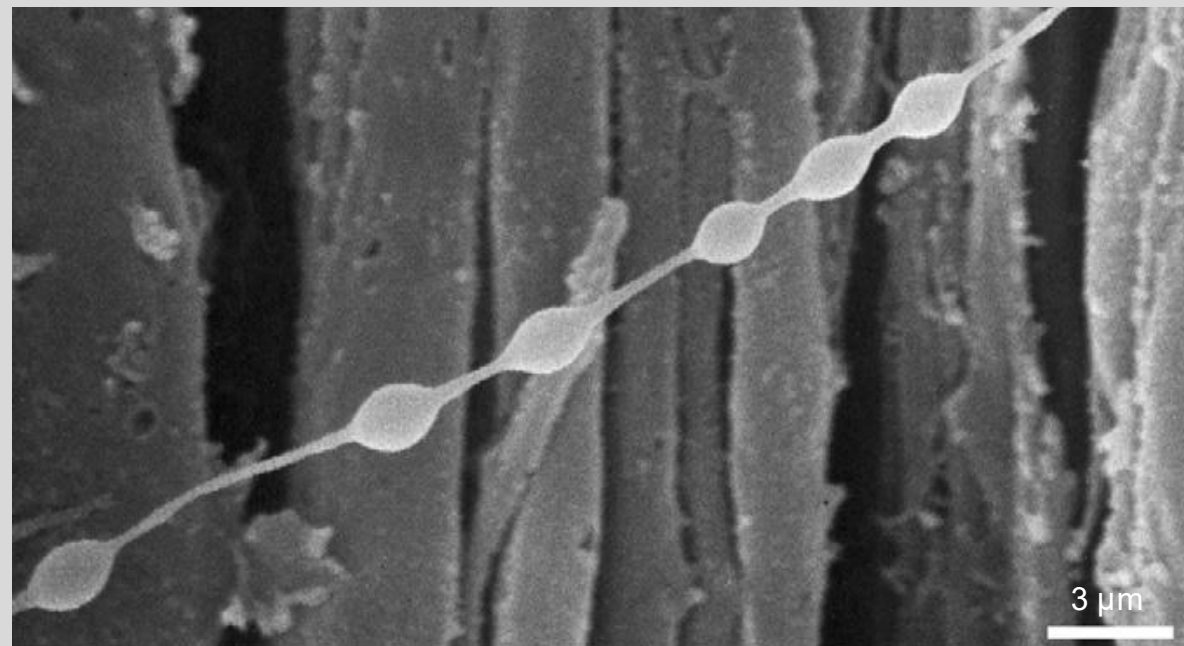
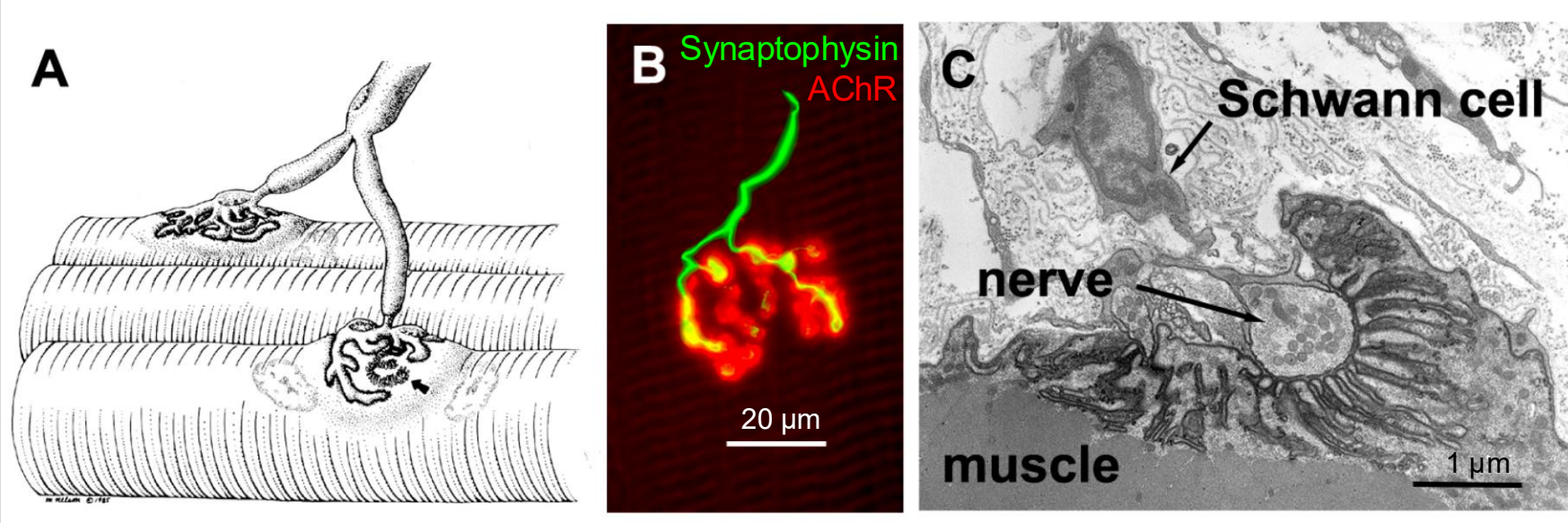
Trends in Neurosciences, 2009-05-01, Volume
32, Issue 5, Pages 241-248

Somatic vs Autonomic Muscle Innervation

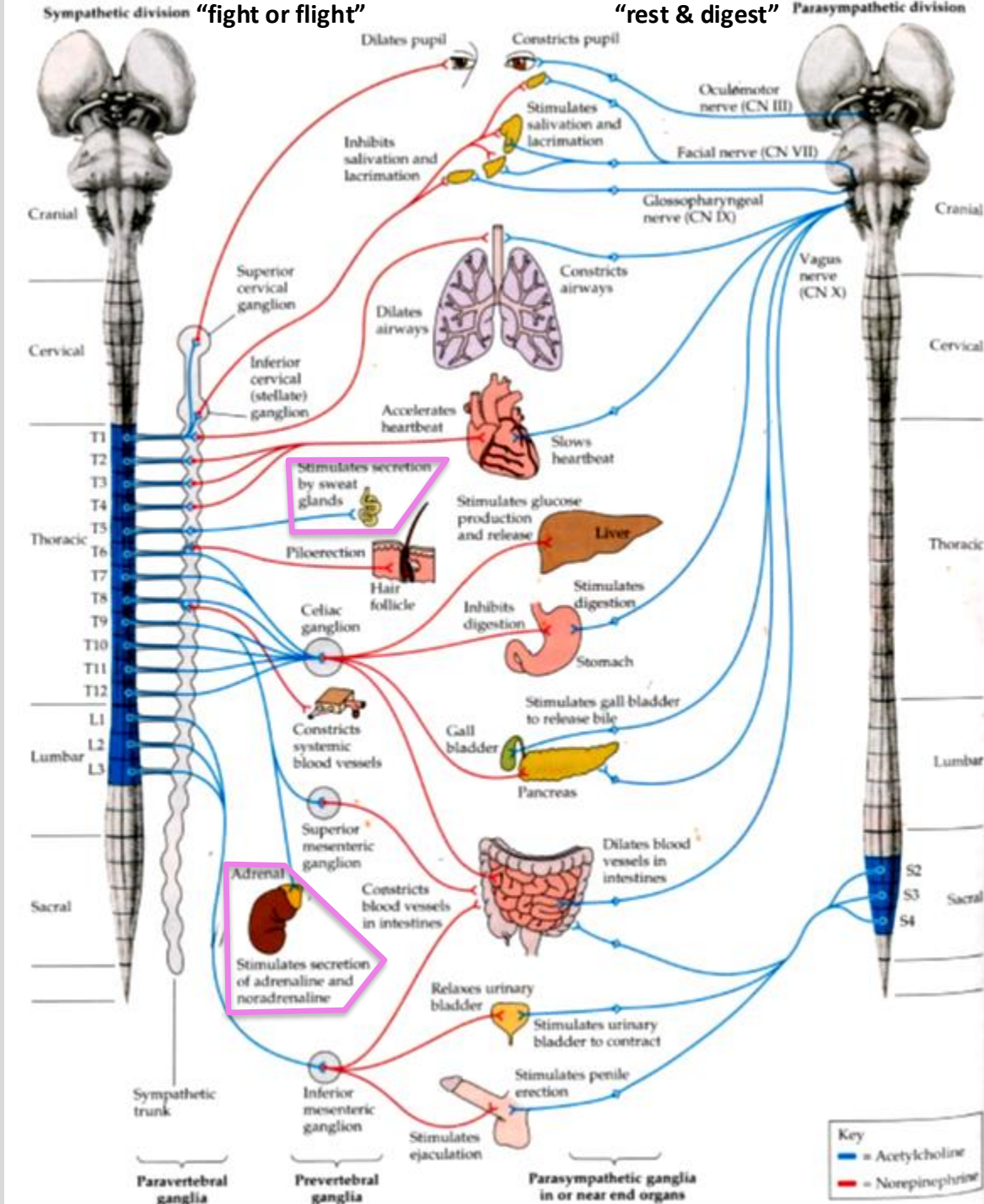
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Int J Mol Sci. 2017; 18(10): 2183.



Neurochem Int. 2008;
52(1-2): 14-25.



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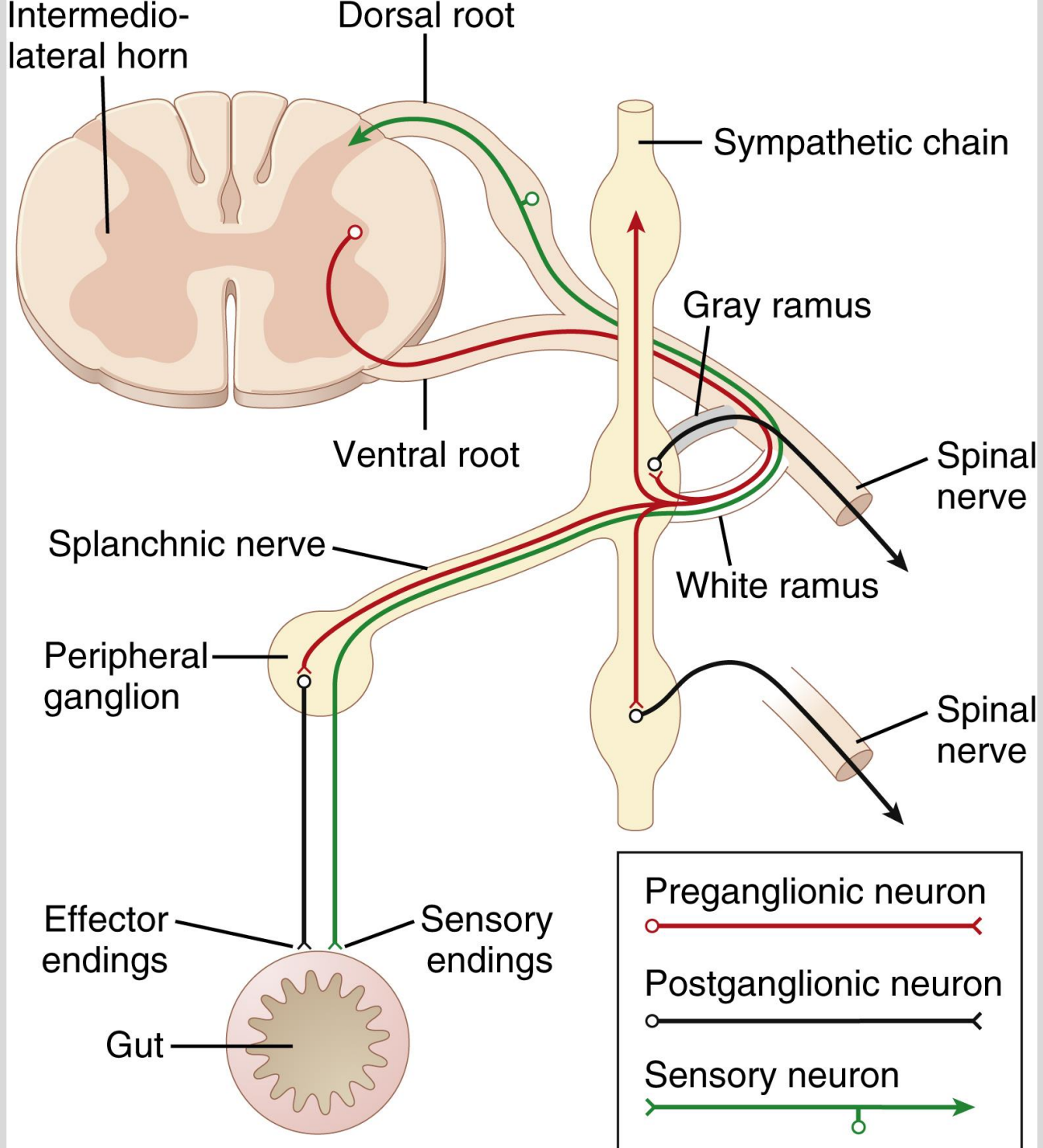
Sweat glands secretions activated by sympathetic postganglionic cholinergic neurons (via M₃).

The adrenal medulla:

- Chromaffin cells – neuroendocrine cells
- Secretes epinephrine (80%) and norepinephrine into the bloodstream
- Sympathetic effect sustained

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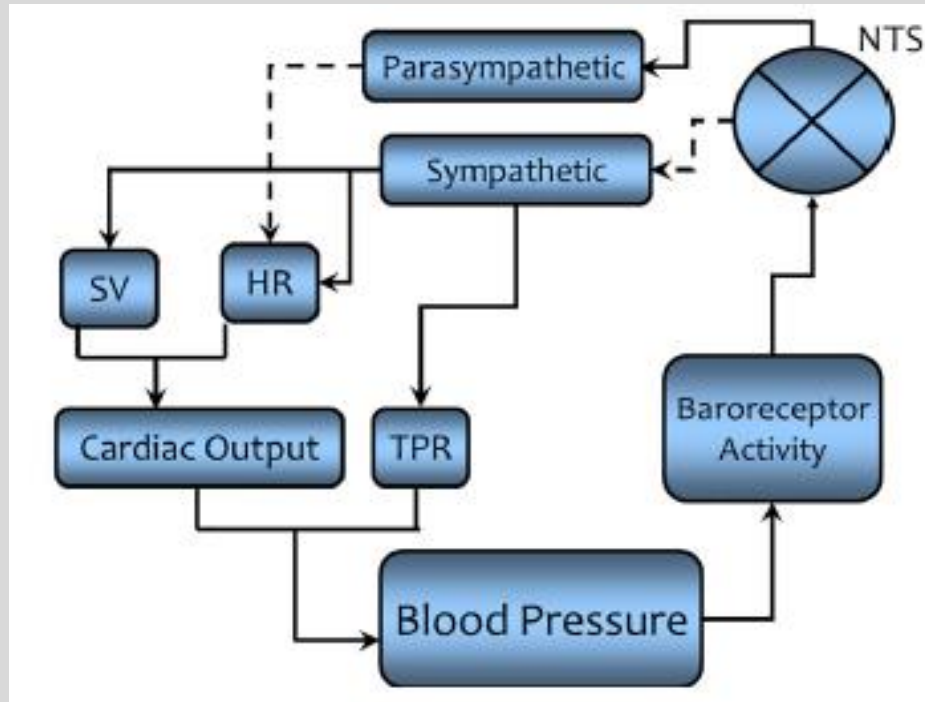


Guyton and Hall Textbook of Medical
Physiology, Chapter 61, 763-775

Baroreceptor Reflex

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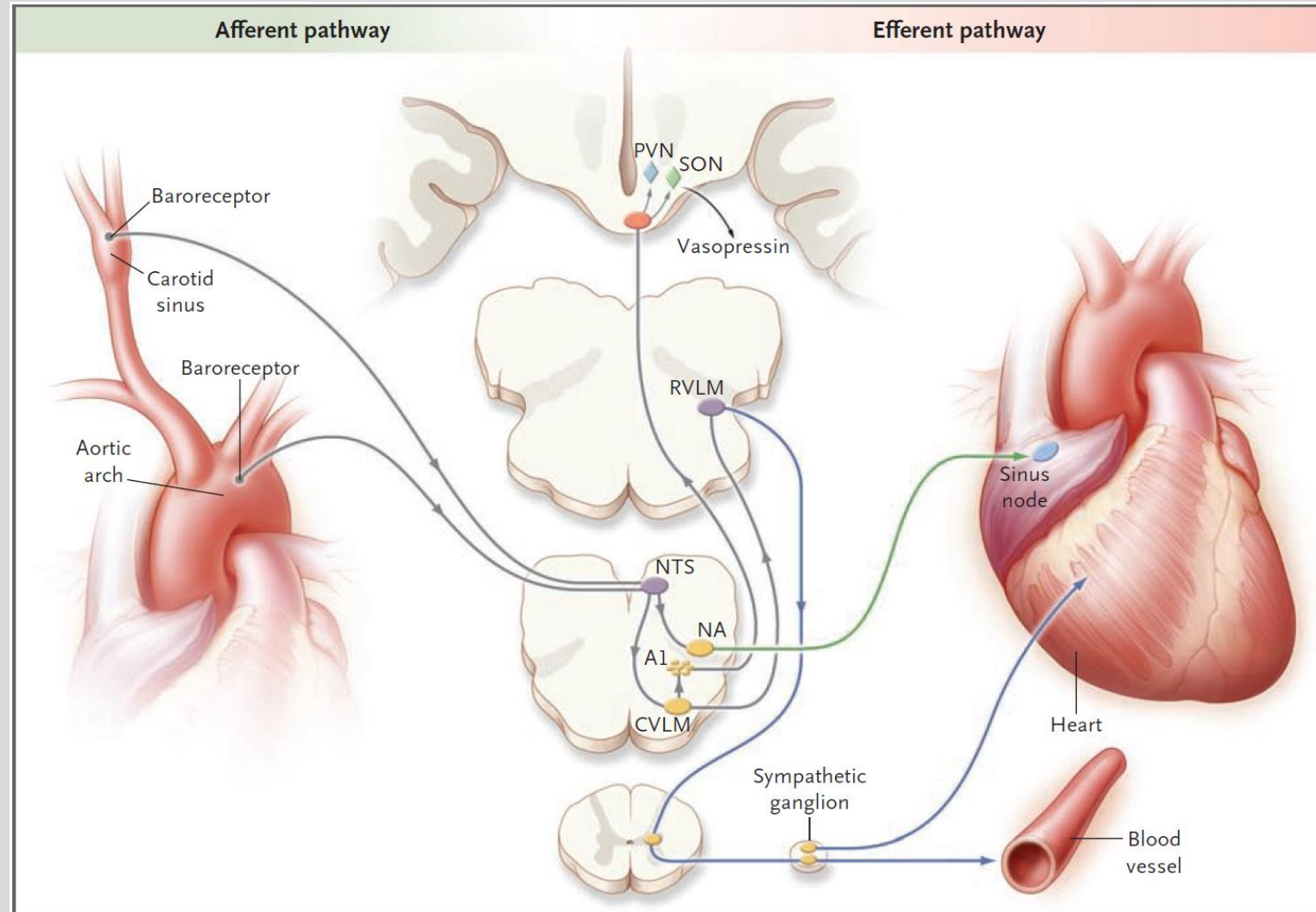
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Front Physiol. 2020; 11: 200.

NTS, the nucleus of the tractus solitarius
NA, the nucleus ambiguus
CVLM, caudal ventrolateral medulla (**inhibitory**)
RVLM, rostral ventrolateral medulla
PVN, the paraventricular nucleus
SON, the supraoptic nucleus

N Engl J Med. 2008; 358(6): 615-24.

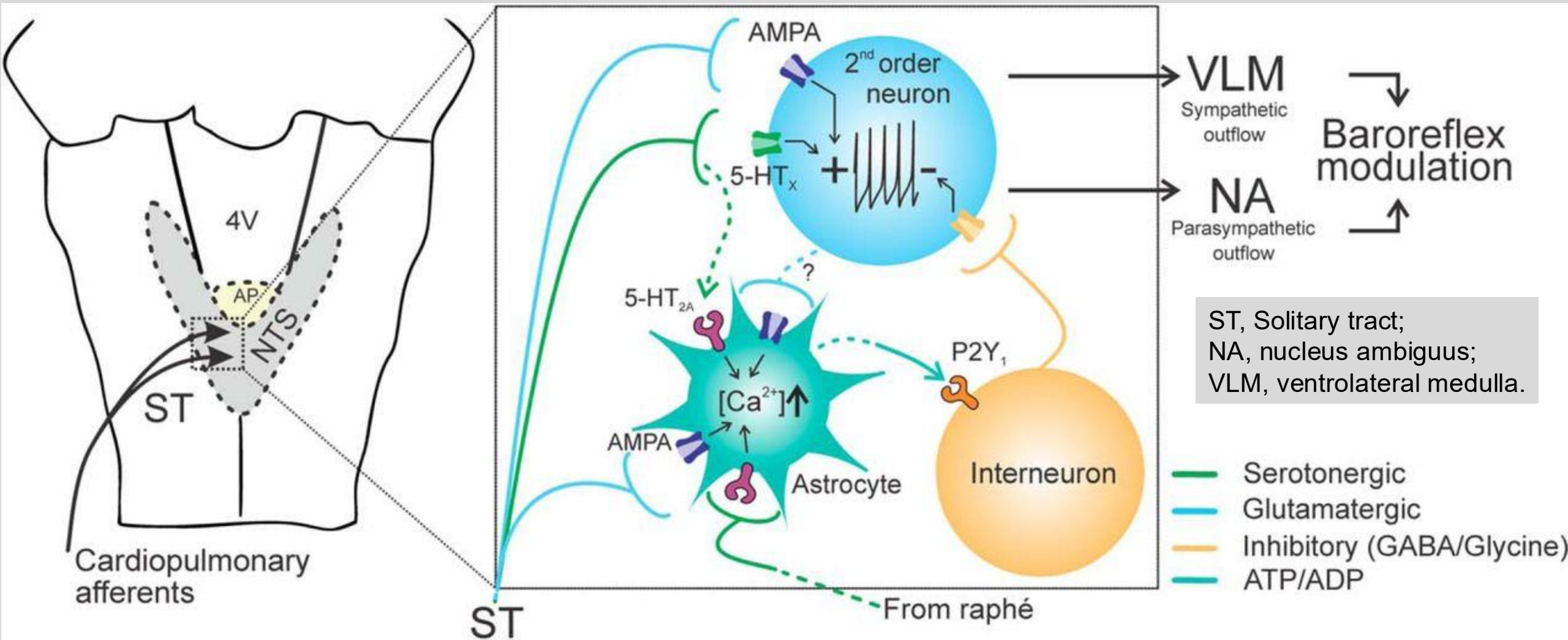


Astrocytes Modulate Baroreflex Sensitivity at the Level of the Nucleus of the Solitary Tract

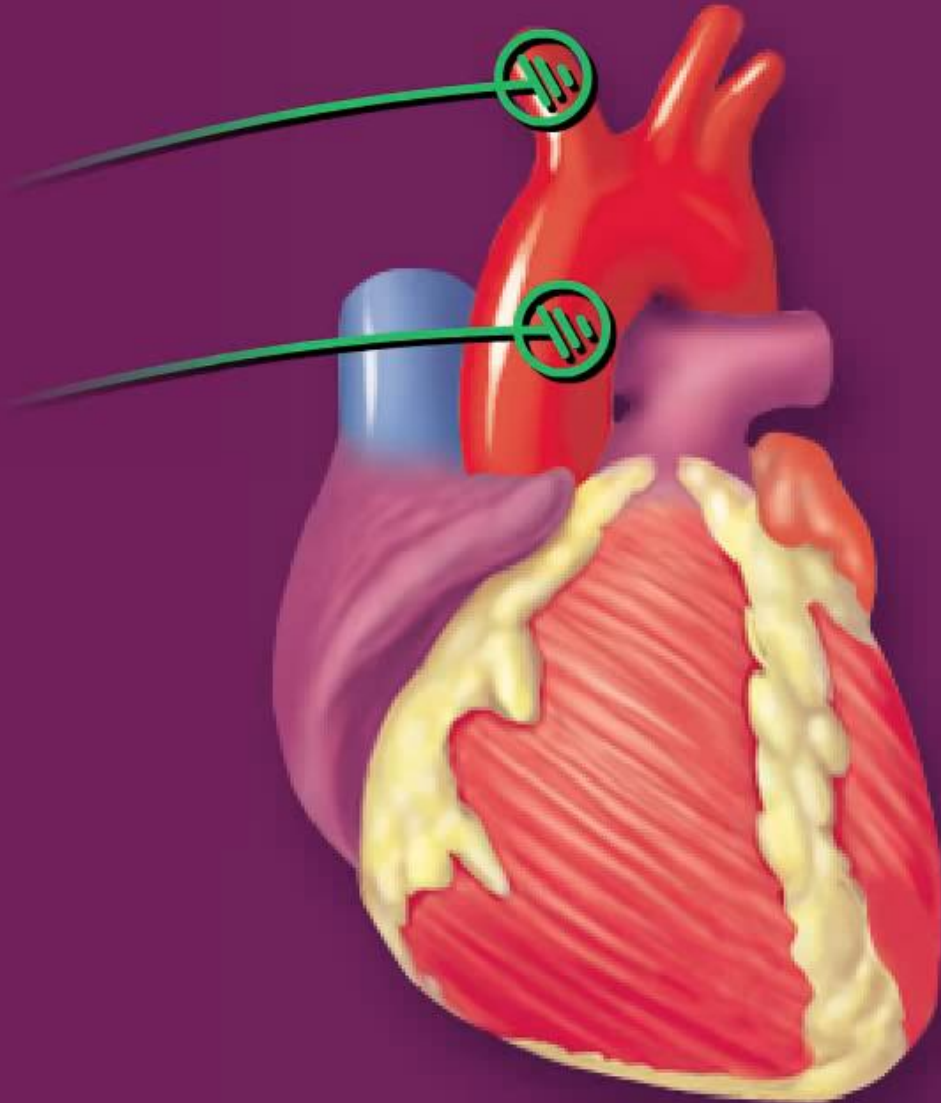
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J Neurosci. 2020; 40 (15): 3052-3062.



Baroreceptor Reflex



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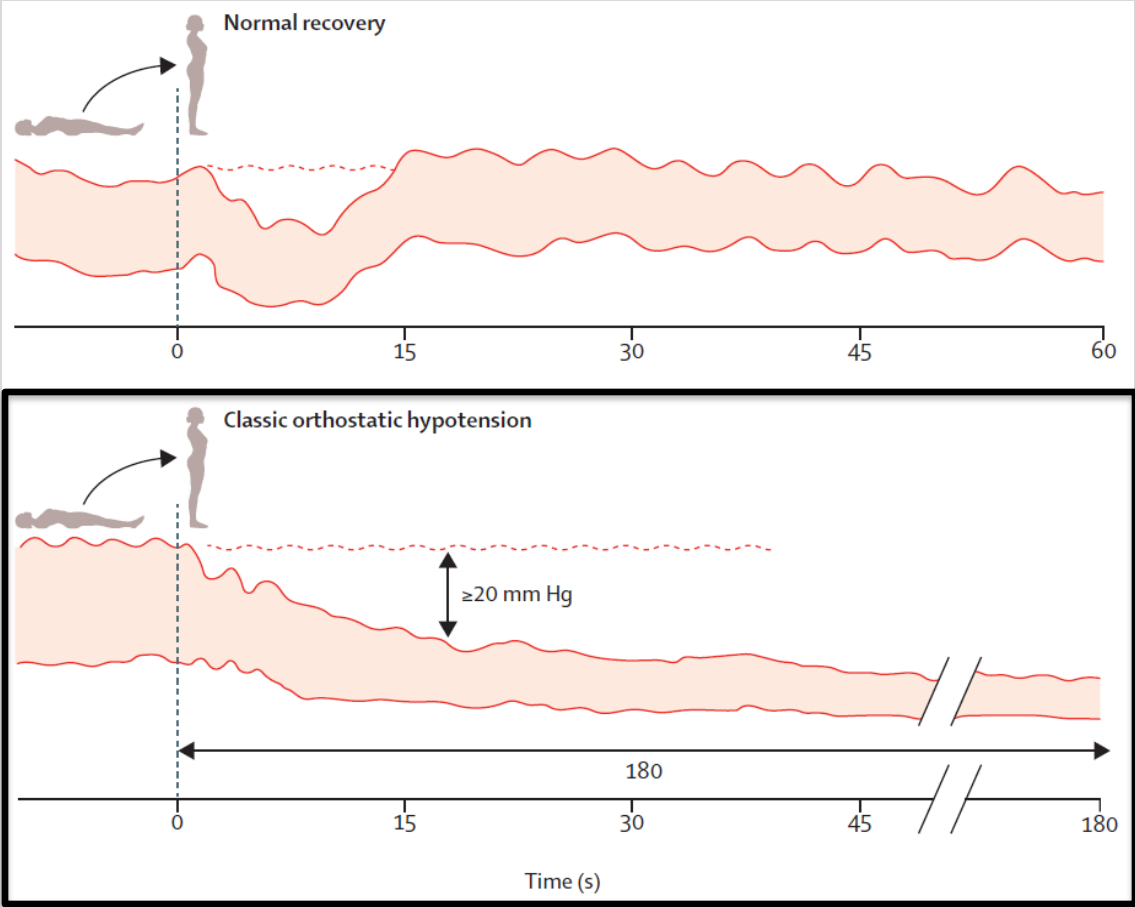
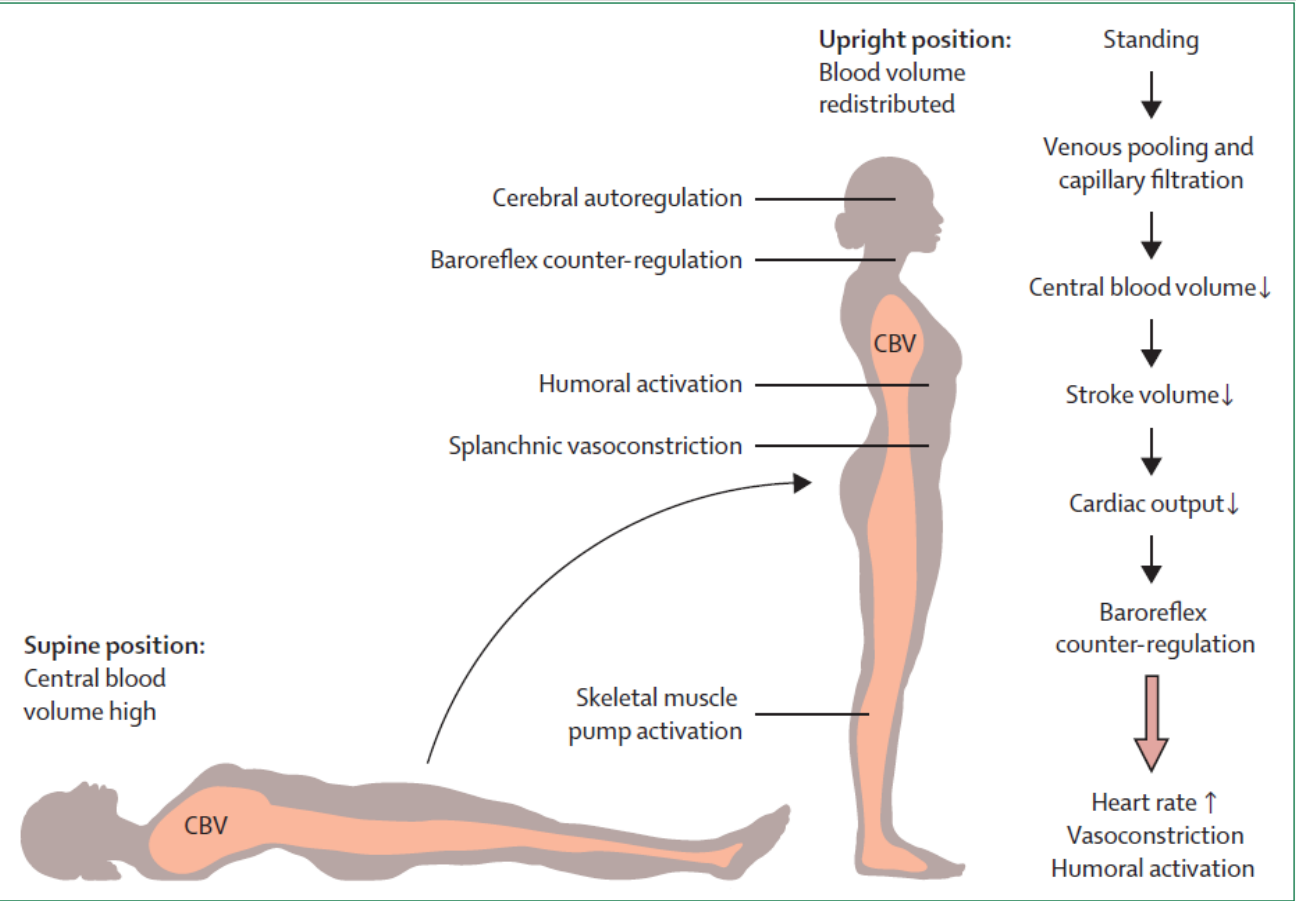
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[https://www.youtube.com/
watch?v=Tr3Nn25TrUU](https://www.youtube.com/watch?v=Tr3Nn25TrUU)

Orthostatic hypotension

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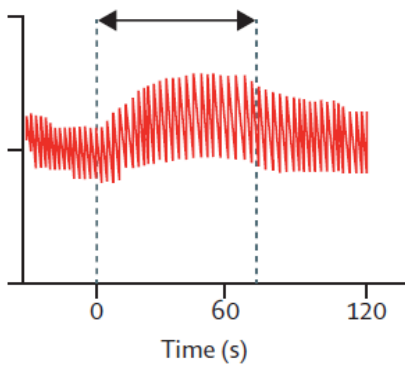
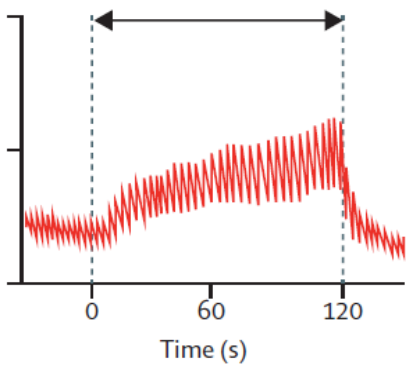
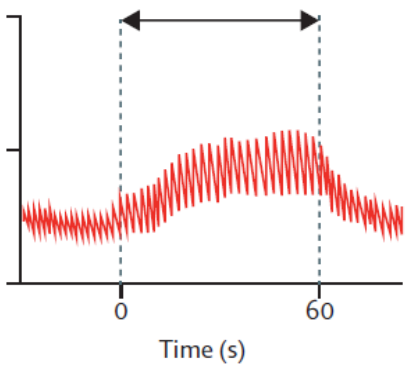
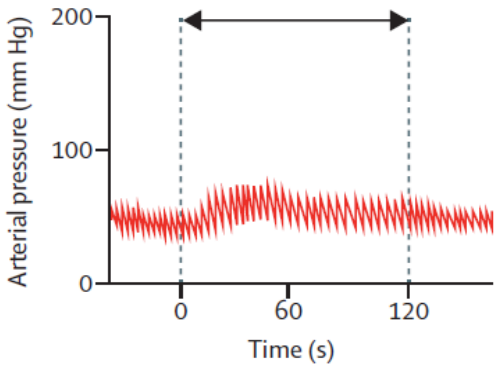
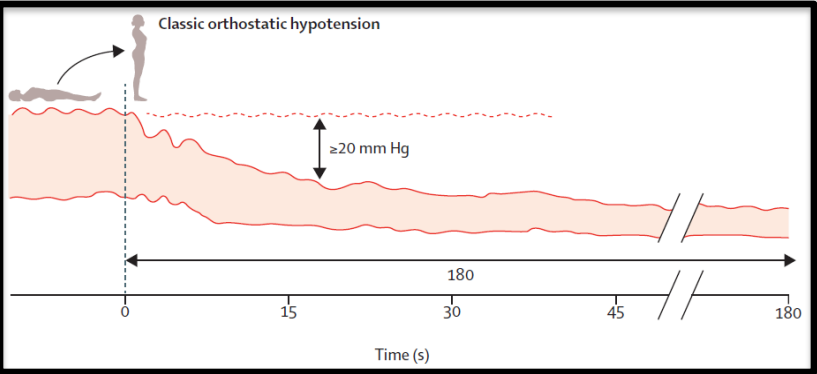
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Orthostatic hypotension

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Carotid Sinus Massage

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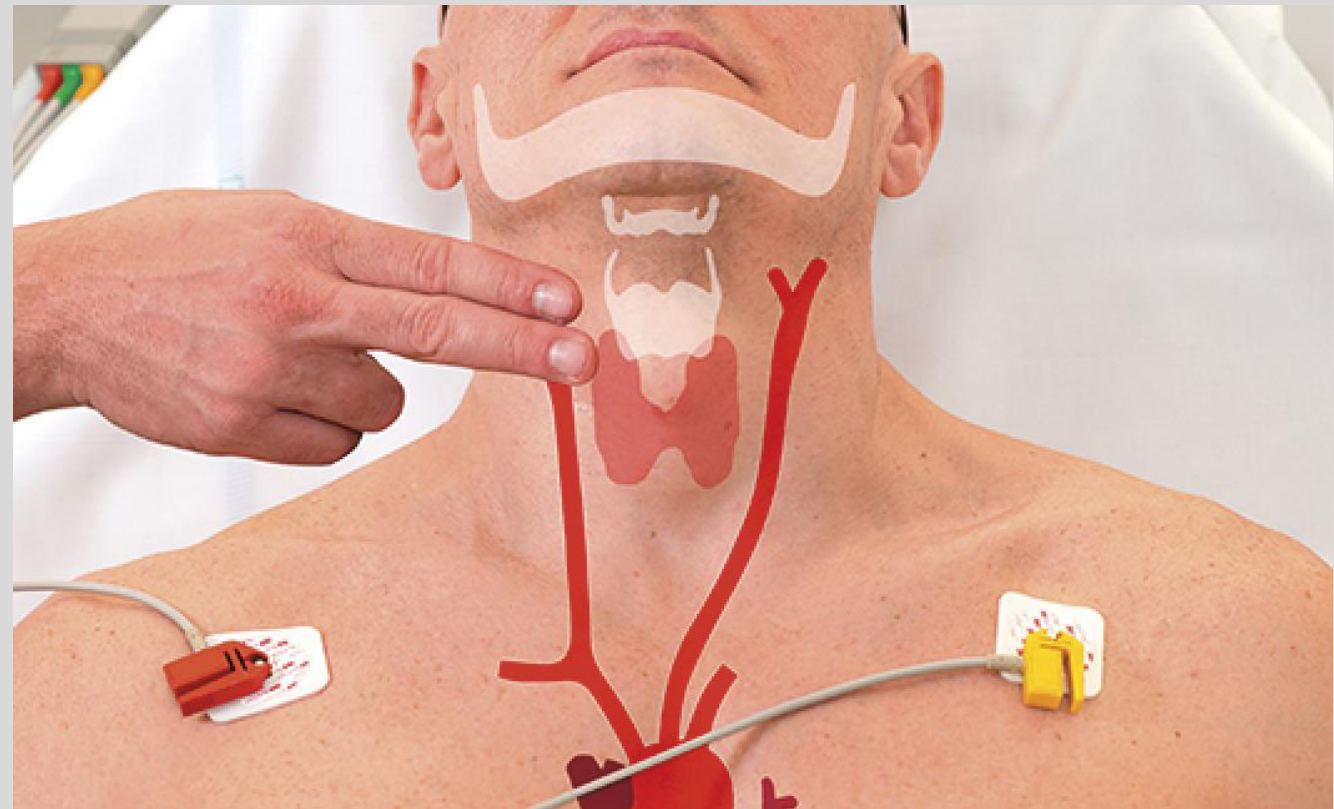
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Indications:

- Initial treatment of hemodynamically stable paroxysmal supraventricular tachycardia
- For the evaluation of syncope in patients older than 40 years of age

Contraindications:

- Patients who are at risk for stroke due to carotid artery disease
- Patients who have had a TIA or stroke within the preceding 3 months or who have clinically significant ipsilateral carotid stenosis or bruit
- Complications from the previous massage, myocardial infarction in the preceding 3 months, or any history of ventricular arrhythmia

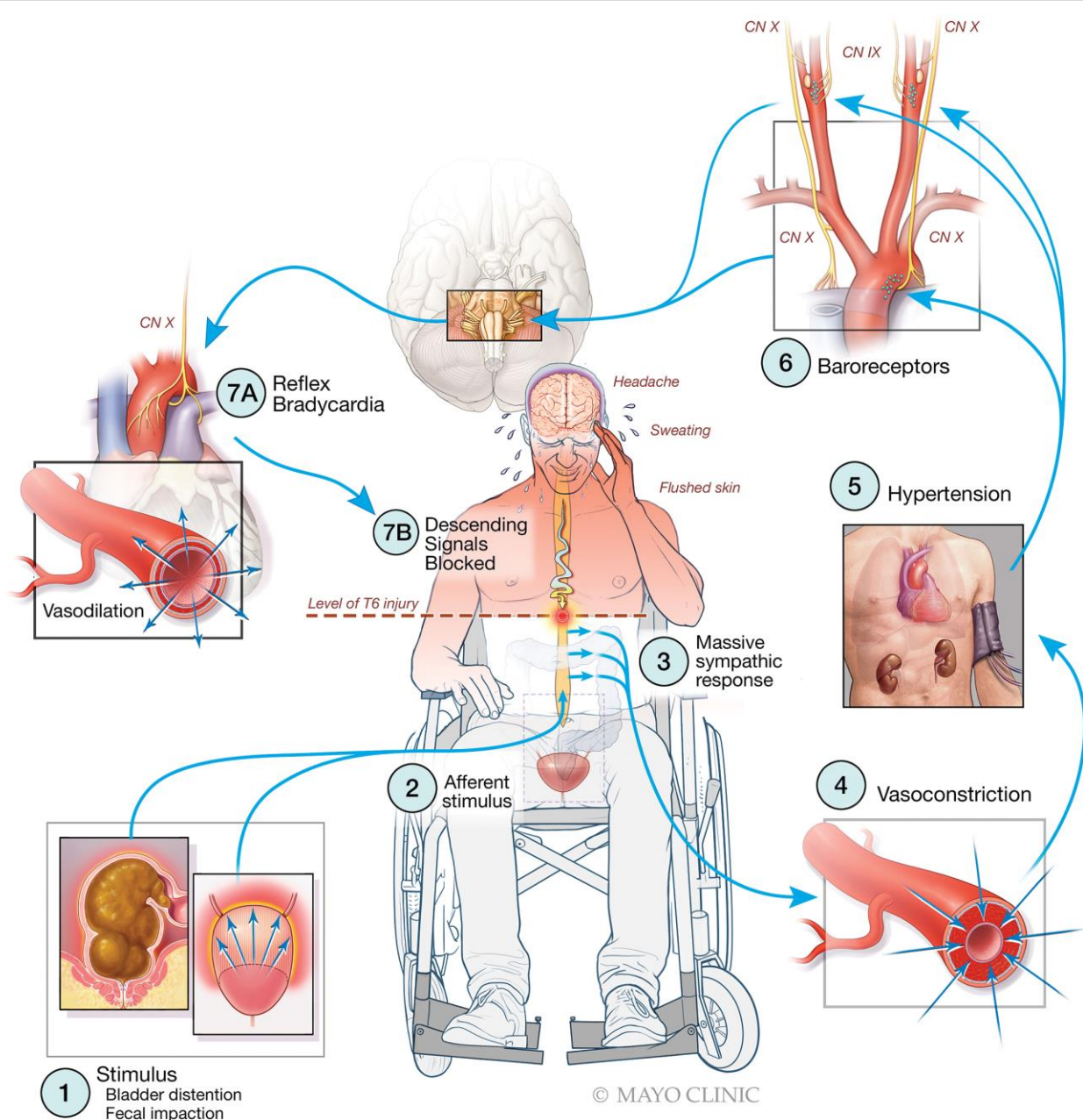


Video: <https://www.nejm.org/doi/full/10.1056/nejmvcm1313338>

Autonomic Dysreflexia

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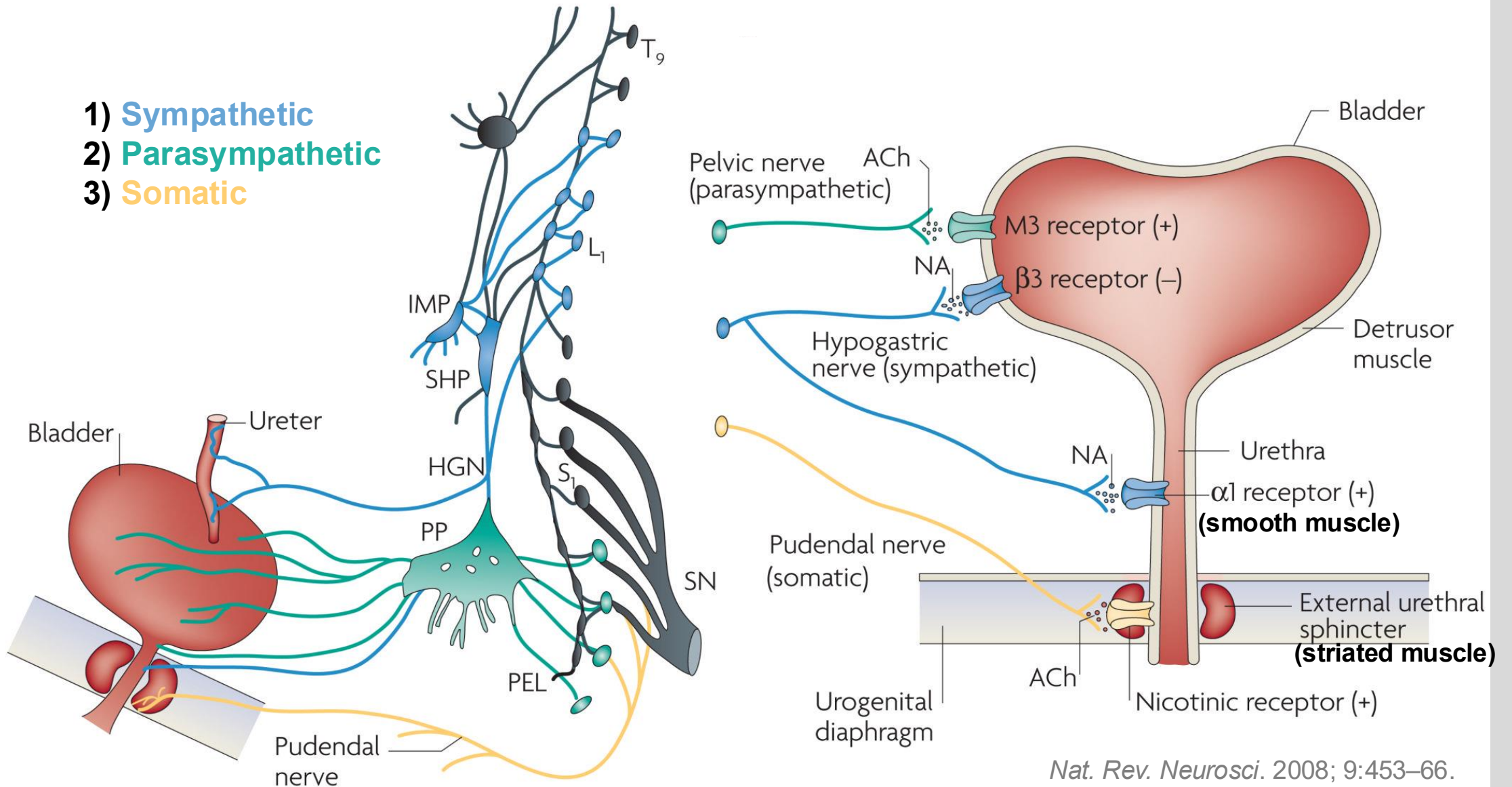


- Consequent to **spinal cord injury**, typically **above T6**
- Hyper-sensitivity of autonomic reflexes to sensory stimuli (i.e. **autonomic spasticity**)
- Sympathetic surge causing widespread vasoconstriction → peripheral arterial hypertension → detected by baroreceptors in the neck → parasympathetic slow heart rate
- **Uncontrolled hypertension and bradycardia!**
- Life-threatening

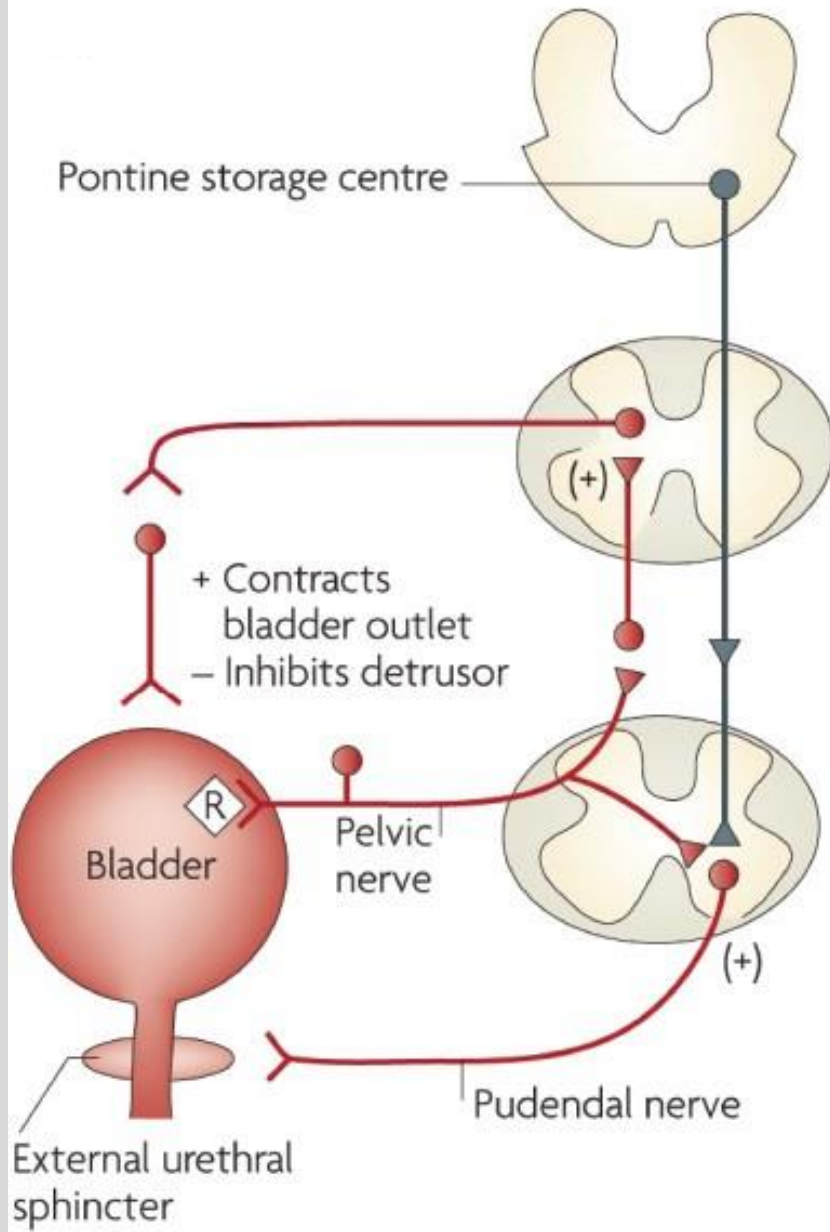
Efferent innervation of the lower urinary tract

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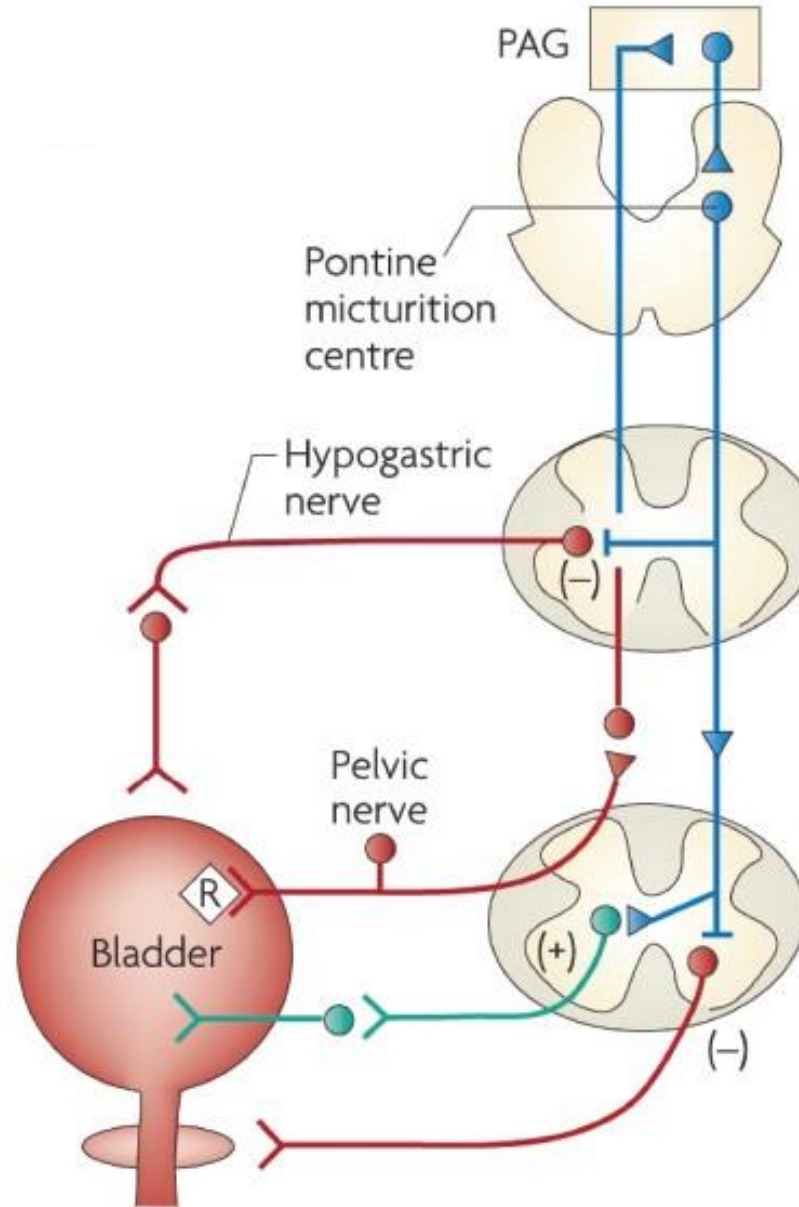
- 1) Sympathetic
- 2) Parasympathetic
- 3) Somatic



Urine storage reflexes



Voiding reflexes

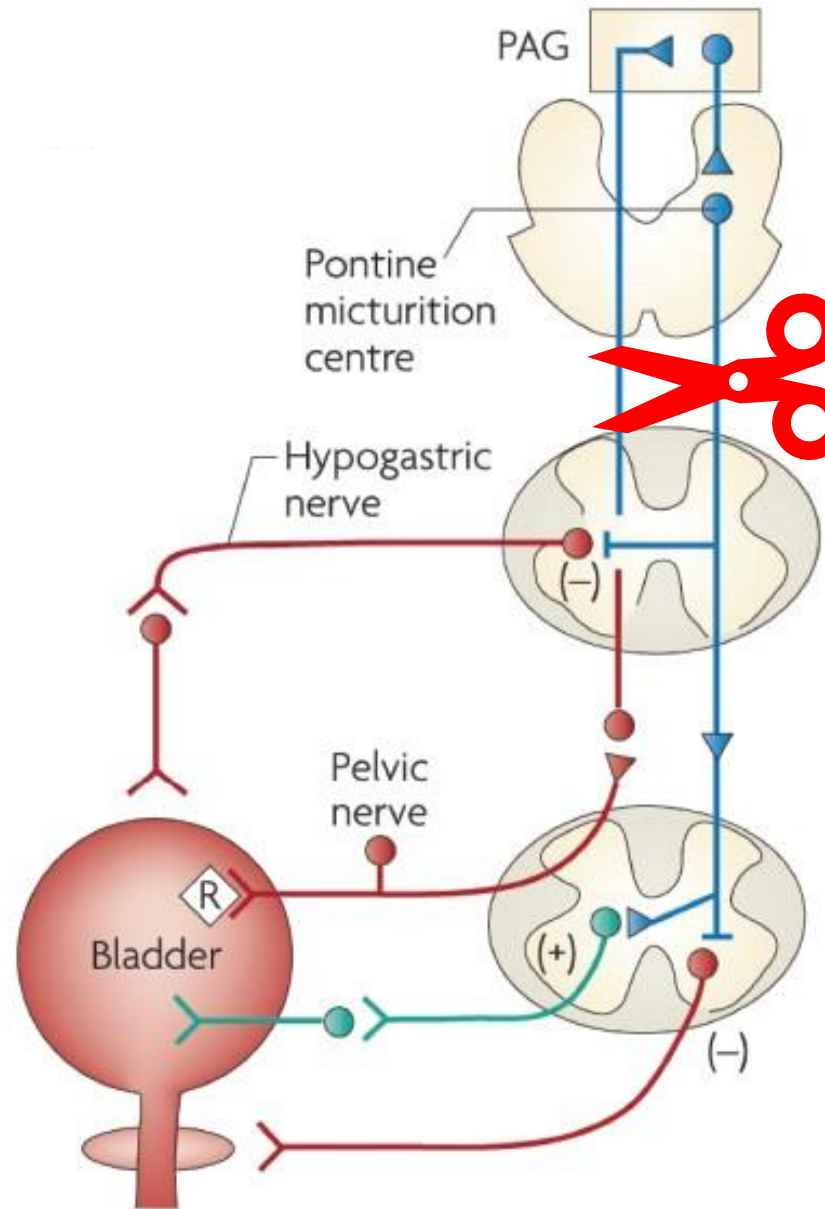
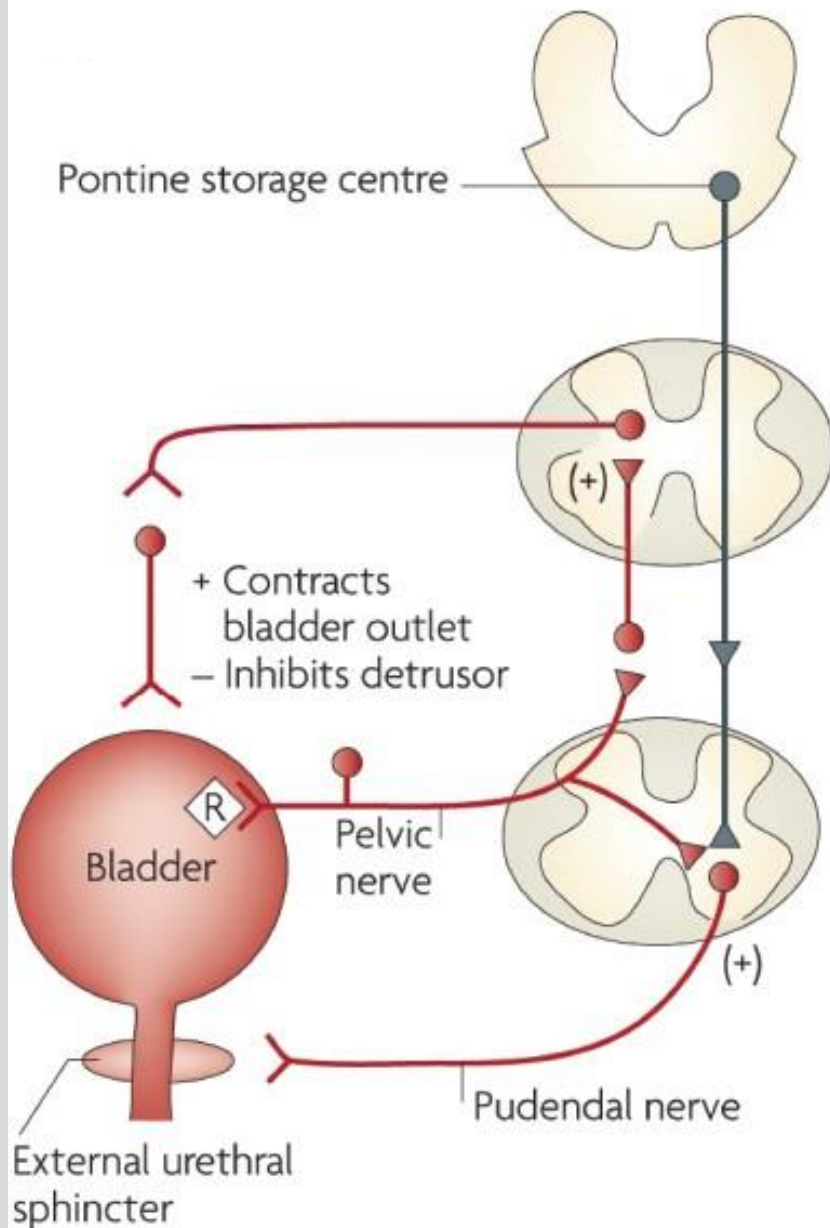


Nat. Rev. Neurosci. 2008; 9:453–66.

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Spastic bladder / Upper Motor Neuron bladder



Nat. Rev. Neurosci. 2008; 9:453–66.

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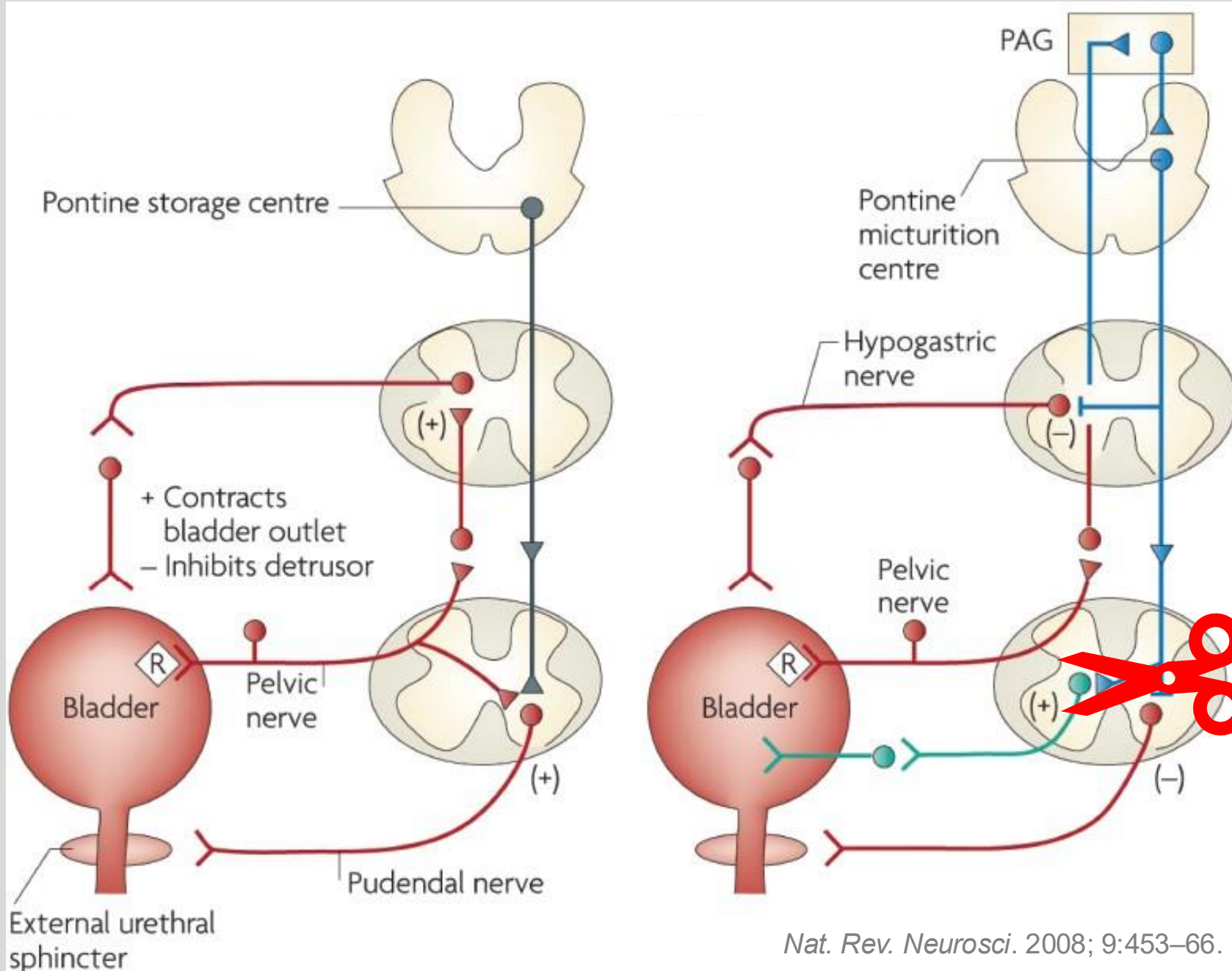
Damage **between the midthoracic spinal cord and the pontine micturition center.**

Vesicovesical reflex reemerges and becomes hyperactive

External sphincter never gets a signal from the **pontine micturition center** telling it to relax → stay contracted

Detrusor has to **contract really hard** to produce enough pressure to overcome the contracted sphincter → urine can back up into the ureters to the kidneys

Atonic bladder / Lower Motor Neuron bladder



Nat. Rev. Neurosci. 2008; 9:453–66.

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Lesions of the **conus medullaris**, **cauda equina** or **sacral cord** are a lot like LMN damage:

Detrusor is unable to contract → bladder expands under relatively low pressure → bladder fills up but never completely empties

External sphincters not contracted → small amount of urine may slowly leak out

The enteric nervous system (ENS)

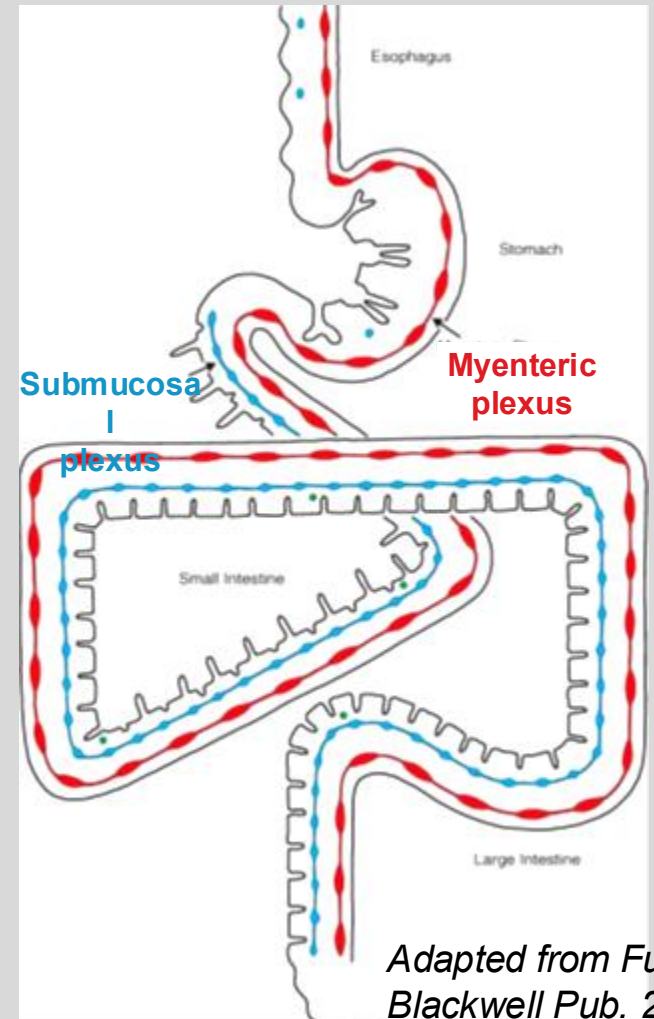
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Enteric neurons

Spinal innervation
Vagal innervation

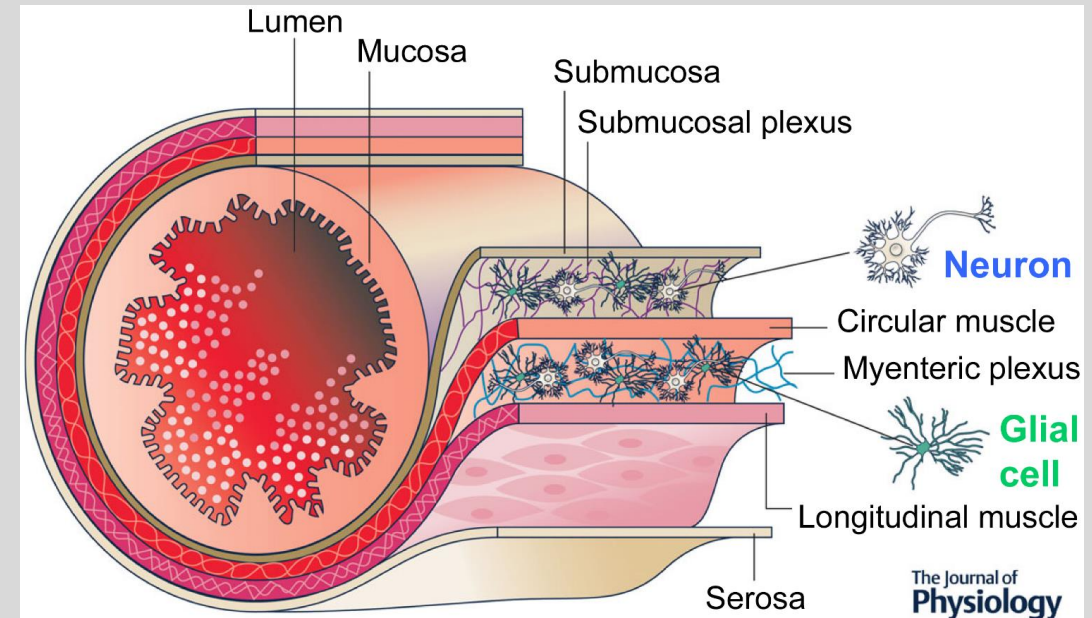
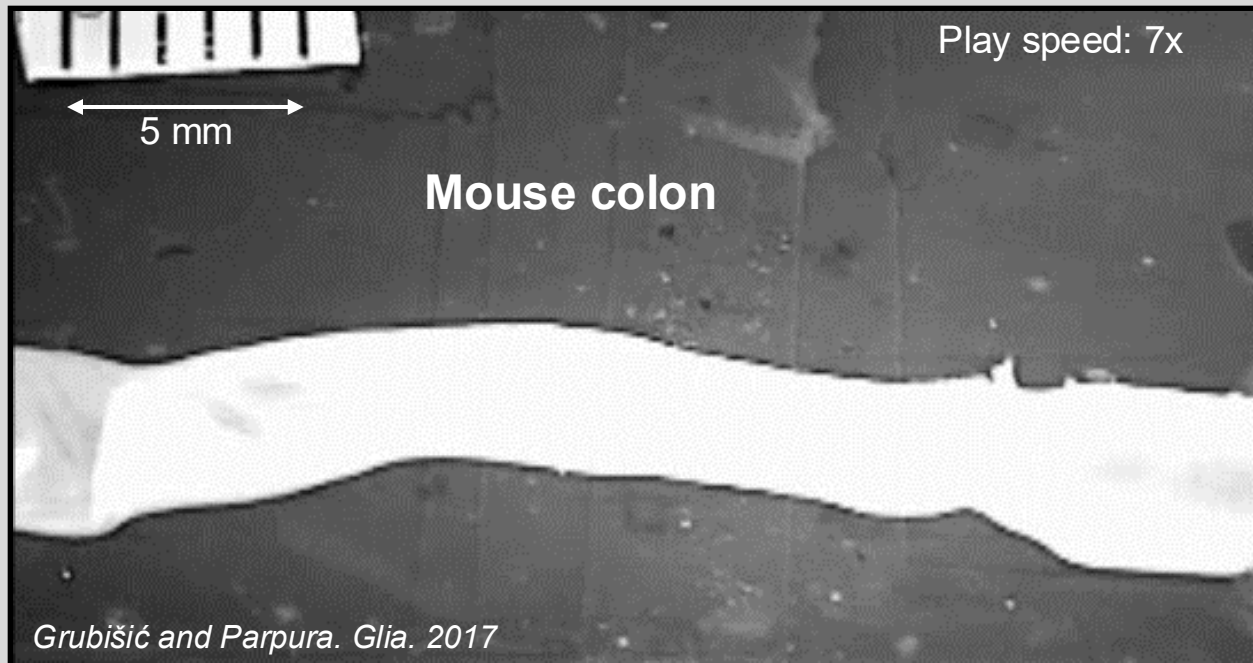
Xueyuan Niu. Cover - JNeurosci. 2020



ENS – 3rd division of the ANS, “the brain in the gut”

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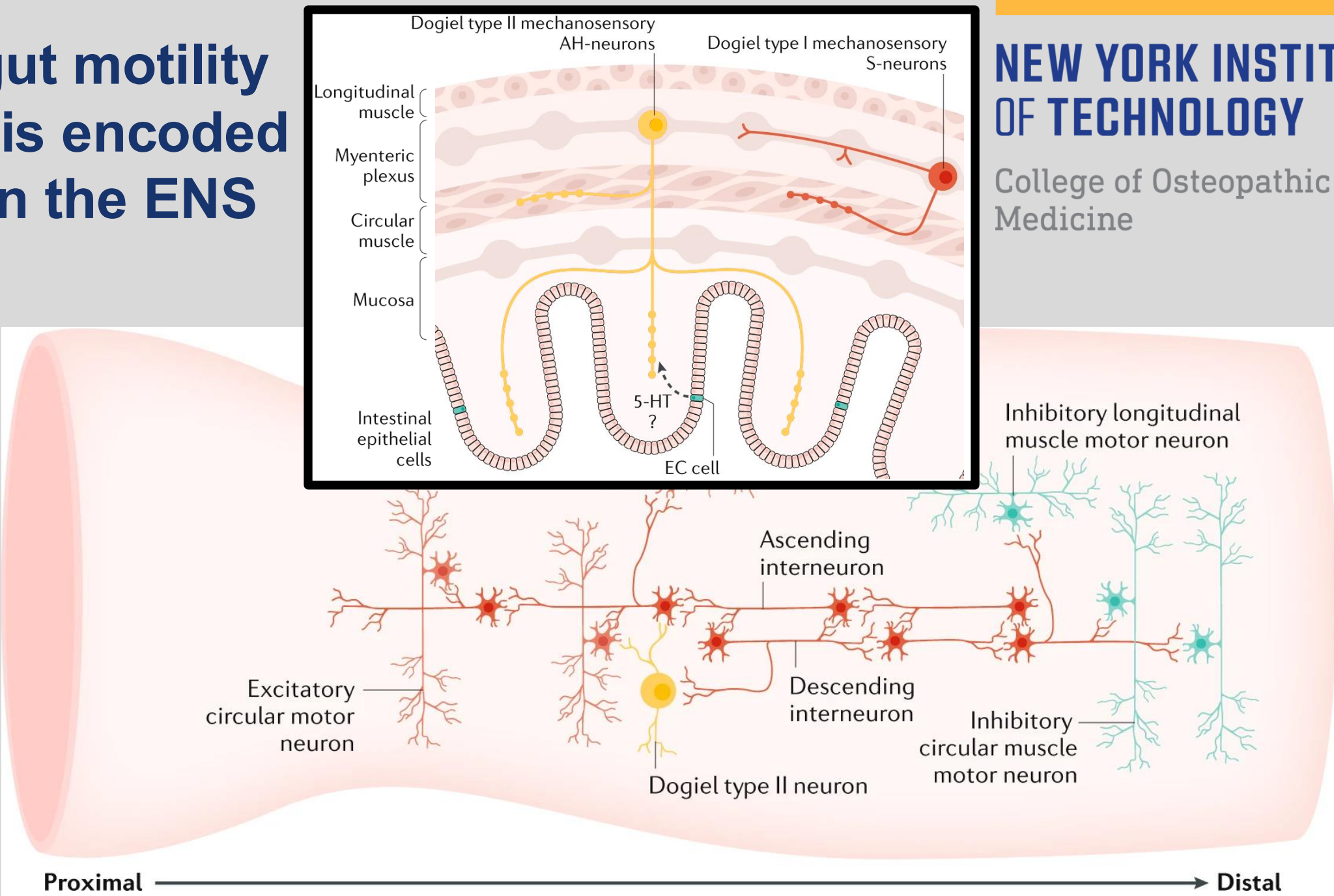


Grubišić and Gulbransen. J. Physiol. 2017.

The gut motility reflex is encoded within the ENS

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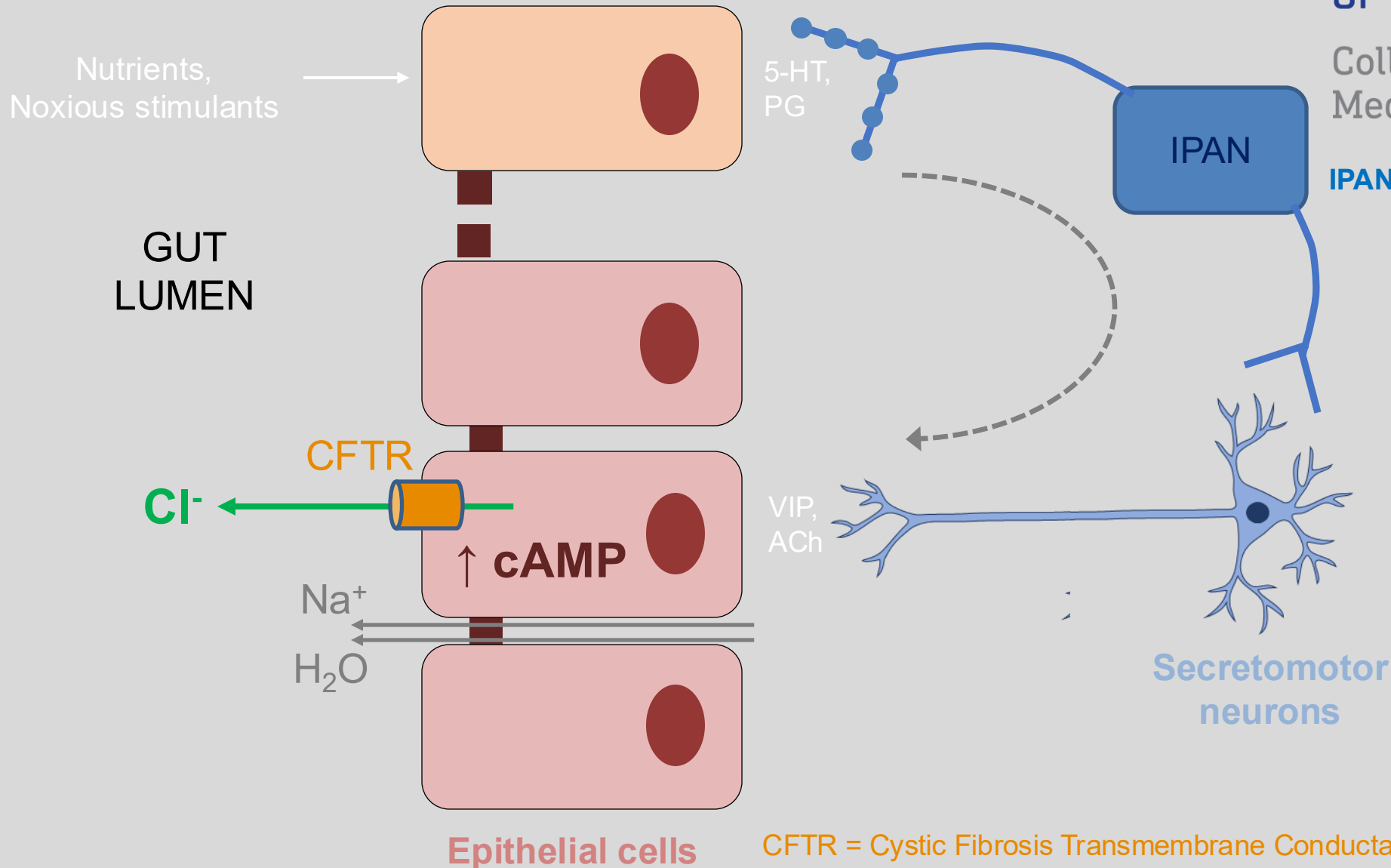
Modified from
Spencer and Hu.
*Nat Rev
Gastroenterol
Hepatol.* 2020.

Secretomotor reflex

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IPAN = Intrinsic Primary Afferent Neuron



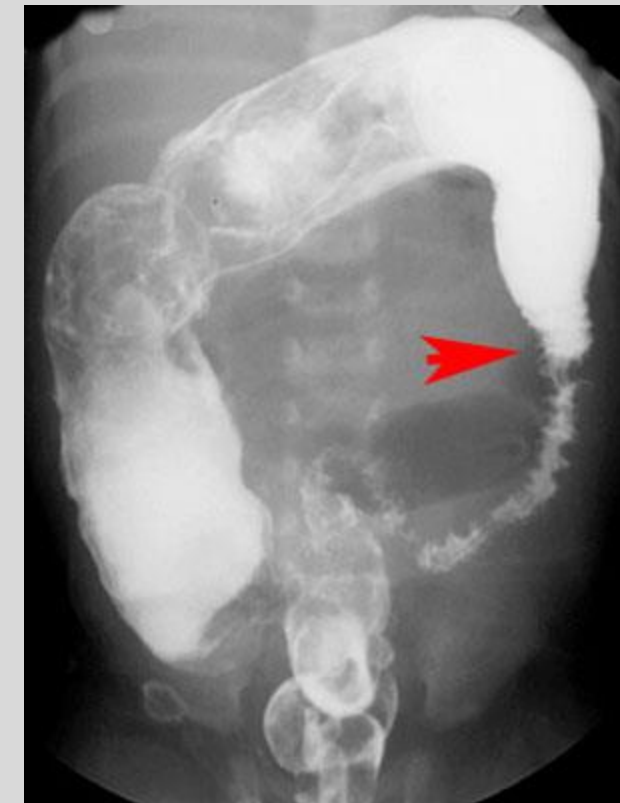
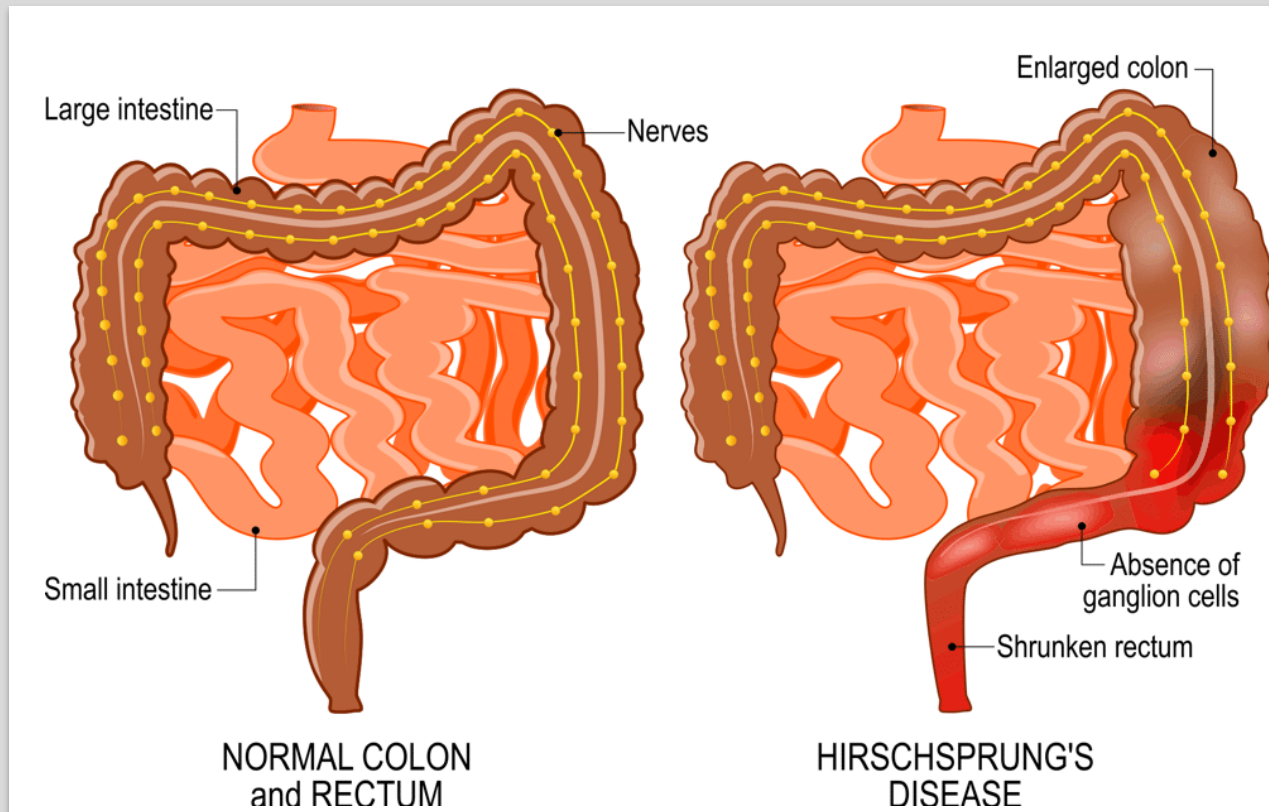
Hirschsprung's Disease (AKA congenital aganglionic megacolon)

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Lack of ENS innervation of the lower gut

- Failure for plexus precursor cells (neural crest cells) to migrate
- Inability to create peristaltic contraction – severe constipation
- Typically identified in newborns (within 24 hrs), requires resection of aganglionic portion



<https://teachmeanipediatics.com/surgery/abdominal/hirschsprungs-disease/>

<https://introductiontoradiology.net/courses/radiology/colon/hirschsprungs.jpg>

Summary

1. ANS and Homeostasis
2. “Rest and digest” (Parasympathetic) vs “Fight or Flight” (Sympathetic)
3. ANS centers in the brain stem and hypothalamus
4. Similarities and differences between the Sympathetic and Parasympathetic divisions
5. Actions of the muscarinic and adrenergic receptors
6. Somatic vs autonomic muscle innervation
7. Baroreceptor reflex
8. Orthostatic hypotension
9. Carotid Sinus Massage
10. Autonomic Dysreflexia
11. Innervation and reflexes of the urinary system
12. Spastic and atonic bladder
13. ENS networks: Myenteric (Auerbach’s) and Submucosal (Meissner’s)
14. Gut motility and secretomotor reflexes
15. Hirschsprung's Disease

Lecture Feedback Form:



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<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>